



Influence of nanoparticle incorporation techniques on tailoring the performance of ceramic nanocomposite membranes for simultaneous heavy metal removal and biofouling control



Somakraj Banerjee, Umme Habiba, Aishwarya Das, Ranjana Das, Chiranjib Bhattacharjee

[Show more](#)

Outline | Share Cite

<https://doi.org/10.1016/j.jece.2026.123031>

[Get rights and content](#)

Full text access

Highlights

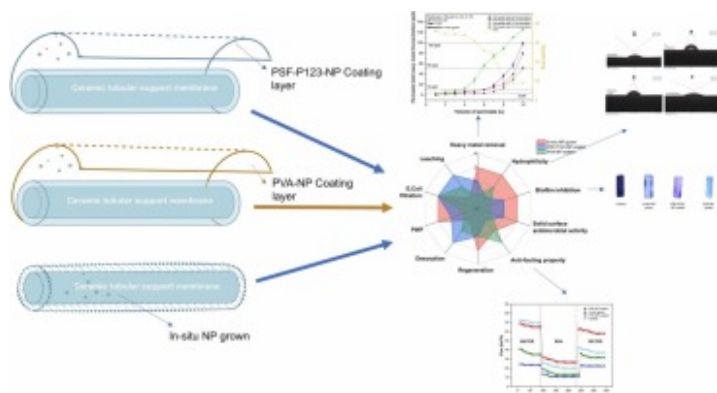
- Nanoparticles were incorporated into ceramic tubular membranes via PSF-P123, PVA coatings, and in-situ growth methods.
- Membranes achieved >96% removal of As, Cd, Pb, Cr; in-situ types projected ~2500L/m² single metal treatment capacity.

- Improved hydrophilicity: water contact angles dropped from 69.83° to 25.06° (in-situ), 38.58° (PVA), and 62.13° (PSF-P123).
- PVA-NP coated membranes exhibited 95.47% flux recovery after filtration of 1000 ppm BSA solution.
- Membranes showed 95–98% antimicrobial activity, 80–83.52% biofilm resistance, and >99.99% CFU removal rates.

Abstract

Mesoporous Mg-Al-Ti ternary composite oxide nanoparticles (NP) were incorporated into ceramic tubular membranes (length: 10 cm, external diameter: 5 mm) via three different fabrication techniques: polysulfone (PSF)-P123-NP coating, polyvinyl alcohol (PVA)-NP coating, and in-situ NP growth within membrane pores. These ceramic nanocomposite membranes demonstrated promising multi-metal removal for both cationic and anionic species across the pH range 3–10, excellent anti-fouling properties, and enhanced antimicrobial capabilities. The in-situ NP grown membrane achieved a significantly high heavy metal removal, treating ~6 litres (~2500 L without regeneration, if scaled up to 1 m² membrane) of 1 mg/l single-metal contaminated water and ~4 litres of multi-metal contaminated wastewater with >96% removal efficiency for As, Cd, Pb, and Cr simultaneously. Hydrophilicity significantly improved, with water contact angles decreasing from 69.83° (unmodified) to 25.06°, 38.58°, and 62.13° for in-situ grown, PVA-coated, and PSF-P123 coated membranes, respectively. PVA-coated membranes exhibited superior anti-fouling performance with 95.47% flux recovery after permeating 1000 mg/l bovine serum albumin. In-situ grown and PSF-P123 coated membranes demonstrated comparatively better antimicrobial properties, including larger inhibition zones (22–24 mm), higher solid-surface antimicrobial activity (95–98%), and stronger biofilm resistance (80–83.52%). All membranes achieved >99.99% removal of colony-forming units (CFU/ml) within 30 min of filtration. These comprehensive results indicate the multifunctionality of the novel nanocomposite membranes and the role of nanoparticle incorporation behind those properties.

Graphical Abstract



[Download: Download high-res image \(159KB\)](#)

[Download: Download full-size image](#)



Keywords

Ceramic; Nanocomposite; Adsorption; Anti-microbial; Heavy metal; Anti-fouling

1. Introduction

The contamination of groundwater and surface water by heavy metals remains a critical environmental issue in contemporary times. Beyond compromising potable water sources, heavy metals tend to bioaccumulate in aquatic systems, adversely affecting aquatic organisms and leading to secondary contamination of soil environments [1]. Regulatory bodies like the United States Environmental Protection Agency (EPA), the World Health Organisation (WHO), and the Central Pollution Control Board (CPCB) of India have identified several heavy metals—namely arsenic (As), chromium (Cr), cobalt (Co), nickel (Ni), copper (Cu), zinc (Zn), cadmium (Cd), mercury (Hg), thallium (Tl), selenium (Se), and lead (Pb)—as priority pollutants [2], [3]. In the present study,

attention is directed toward the removal of As, Pb, Cd, and Cr, which are among the most hazardous. The maximum permissible concentrations for these contaminants, as specified by WHO and CPCB, are 10 µg/L, 10 µg/L, 3 µg/L, and 50 µg/L, respectively [4], [5]. Techniques such as coagulation, flocculation, adsorption, membrane separation, ion exchange, and electrochemical processes have been investigated extensively [6], [7], [8]. Membrane-based separations—including nanofiltration and reverse/forward osmosis—are considered viable; however, they are often hindered by membrane fouling and elevated operational costs [9]. Nanostructured adsorbents provide enhanced surface areas for metal ion binding but suffer from drawbacks such as agglomeration, low mechanical strength, and challenges in post-treatment separation [10]. The limitations of traditional membrane separation and adsorption using nanomaterials can be mitigated by incorporating nanoparticles into membranes by immobilising nanomaterials within the membrane matrix, giving properties such as hydrophilicity, mechanical stability, anti-fouling behaviour, and microbial resistance [11], [12].

Studies on combining ceramic membranes with functional nanomaterials have been going on for some time. Wu *et al.* used a non-solvent-induced phase separation-spin coating (NIPS-SC) strategy to coat ZIF-8 metal-organic framework on the membrane surface for enhanced water flux and contaminant rejection [13]. Saleh *et al.* grafted ZnO nanorods on a basalt ceramic membrane by the solution deposition method, which showed enhanced antimicrobial properties and some self-cleaning properties [14]. Menne *et al.* fabricated a polymer/ceramic hybrid nanofiltration membrane based on polyelectrolyte assembly by the layer-by-layer technique, which made the composite membrane structurally stronger and regenerable [15]. Maaskant *et al.* fabricated a thin polyamide layer on a thin, porous hollow-alumina fibre by interfacial polymerisation of piperazine with trimesoyl chloride [16]. Hamm *et al.* developed an innovative method of dip coating alumina ceramic membranes in polymer solution, followed by phase inversion to fabricate a polymer-ceramic composite membrane, but the study didn't discuss much about the functionality of such coatings [17]. Although these composite membranes have been prepared with different and arguably complicated fabrication techniques, these membranes lacked the multifunctionality that could be achieved by adding more layers to a ceramic membrane. It is also important to note that most of the modifications on the ceramic membranes have been surface-level modifications and not membrane matrix-level modifications. Bai *et al.* tried a matrix-level modification to ceramic membranes by adding MgO and Al₂O₃ nanoparticles in the ceramic mould [18]. In this method, due to the high temperature needed to prepare the ceramic membrane, the nanoparticles introduced in the mould may lose some of their functionality, as seen in our previous work [19]. Roy *et al.* developed an amine functionalized composite membrane based on ceramic support, where they fabricated two layers using polyvinyl alcohol, alumina and boehmite, followed by EDTA treatment [20]. It could remove As(III), Cr(VI) and Cu(II) successfully, but the use of high sintering temperature in this method also puts the addition of

functional nanoparticles in question. Barati *et al.* tried in-situ growth of iron nanoparticles in ceramic support, but the study doesn't delve into functional aspects of the membrane [21]. Although there are multifunctional nanocomposite polymer-based membranes showing heavy metal removal, anti-fouling and anti-microbial properties, there have been very few studies to prepare nanocomposite ceramic membranes with multifunctional properties. Many different fabrication methods have been used with no certain way to compare the efficiency of the fabrication methods. That is why a comparative analysis of different nanoparticle incorporation methods using the same nanoparticle comes to be a necessary research work to fill the knowledge gap

Therefore, this work attempts to address the knowledge gap by finding inferences on comparative analysis of membrane performances, which will allow fine-tuning and proper designing of ceramic nanocomposite membranes. Adding to the previous state of the art, this work also presents three new ceramic nanocomposite membranes, adding to the technological knowledge of ceramic nanocomposite membranes. Mg-Al-Ti composite oxide NP has shown exceptionally high As adsorption capacity (1215 mg/g for As(V), 919 mg/g for As(III)) along with very high adsorption capacity for Pb (1103 mg/g), Cr (240 mg/g), Cd (2144 mg/g) as well [12]. This NP is a very promising material for water treatment, and it was selected as the adsorbent to fabricate the ceramic nanocomposite membranes. In the first membrane, matrix-level addition of nanoparticles was done by growing the nanoparticles inside the pores of the ceramic support membrane through a novel fabrication process. The second membrane was prepared by adding a PSF-P123-NP mixed matrix polymer membrane layer on top of the ceramic membrane using another novel process as a surface-level modification. This membrane is arguably one of the few available ceramic-supported polymer membranes incorporating nanoparticles. The third membrane has been prepared by immobilising the NPs on the ceramic membrane surface using a PVA-based coating. Membrane morphologies and incorporation of NPs were studied using FESEM (Field-emission scanning electron microscopy) and EDX (Energy dispersive X-ray) mapping techniques, and X-ray diffraction (XRD). Improvement in the hydrophilicity parameter from 69.83° to 25.06° was studied using a goniometer. Both single-species removal and multi-species removal for the long term have been studied, along with filtration of real heavy metal-contaminated wastewater to determine the breakthrough volume of the membranes. Results suggest that a 1 m^2 membrane could treat $>2500\text{ L}$ of $1000\text{ }\mu\text{g/l}$ As(III) solution without regeneration and $>3000\text{ L}$ of multi-metal solution consisting of As, Pb, Cd, and Cr at $250\text{ }\mu\text{g/l}$ concentration each without regeneration. Anti-fouling property has been studied using 1000 mg/l BSA as the foulant. Anti-microbial, anti-biofilm and solid-surface antimicrobial activity has been studied using *Escherichia coli* and *Staphylococcus aureus* for all the membranes to determine the best NP incorporation method in terms of anti-microbial resistance. All the membranes showed $>99.99\%$ removal of colony-forming units (CFU/ml) within 30 mins of filtration. This study

aims to introduce two different novel fabrication methods (in-situ incorporation and PSF-P123 coating) and a comparative study to help understand the performance of different ceramic nanocomposite membranes prepared infused with Mg-Al-Ti ternary oxide adsorbent NPs. Ultimately, this study will help create a thorough understanding of the effects of different NP incorporation methods on the properties of ceramic membranes.

2. Materials and methods

2.1. Materials

Polysulfone of average molecular weight 35 kDa, precursors for adsorbent synthesis, aluminium isopropoxide (>98 wt%), titanium isopropoxide (97%), magnesium nitrate hexahydrate ($\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$), and the soft template Pluronic P123 ($M_n = 5800$, $\text{EO}_{20}\text{PO}_{70}\text{EO}_{20}$, EO = Ethylene oxide, PO = Propylene oxide) was purchased from Sigma-Aldrich. Arsenic salt NaAsO_2 was purchased from Loba Chemie Pvt. Ltd. Solvent N, N-Dimethylacetamide (DMAc, 99.5%), HNO_3 (about 70% w/w), NaOH, $\text{Pb}(\text{NO}_3)_2$, $\text{Cd}(\text{NO}_3)_2$, and $\text{K}_2\text{Cr}_2\text{O}_7$ were obtained from Merck. The bacterial stock cultures, *Escherichia coli*- NCIM 2065 and *Staphylococcus aureus* NCIM-2679, were purchased from NCIM, Pune. Mueller Hinton, Luria Bertani, Tryptic Soy Broth, Nutrient Broth and agar agar were procured from HiMedia. Ceramic tubular microporous membranes made of alumina, with a length of 10 cm, external diameter 5 mm, and average pore size $1.5\text{ }\mu\text{m}$ were purchased from Need Innovation, Kolkata. Chemical reagents were used without any purification or treatment. Deionised (DI) water from Sartorius Arium Pro was used for synthesis and other experiments.

2.2. Synthesis of Mg-Al-Ti oxide adsorbent nanoparticles

The adsorbent nanoparticles were synthesised using a templating method, where P123 was used as the structure directing agent, absolute ethanol was used as the solvent and aluminium isopropoxide (>98 wt%), titanium isopropoxide (97%), and magnesium nitrate hexahydrate ($\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$) were used as the precursors of Mg, Al and Ti. The detailed process has been discussed in our previous study [19].

2.3. Fabrication of In-situ NP grown ceramic nanocomposite membrane

Taking a novel approach to in-situ growth of NPs in the membrane pores, the evaporation-induced self-assembly synthesis method has been adapted. 2 g P123 was dissolved in 40 ml of absolute ethanol as the template or structure directing agent for the

NP synthesis, along with 3.2 ml HNO_3 (70% w/w) and three precursor salts for Mg, Al, and Ti in a 3:1:1 molar ratio. Since the yield of the process is already known from our previous studies, the total amount of precursor salt was taken to be 7 mmol to adjust the NP content to be 1% w/w. The solution was stirred on a magnetic stirrer for 6 h or until a homogeneous solution was achieved. Once a homogeneous solution was formed, tubular ceramic membranes were completely immersed in it and were then kept in a hot air oven at 60 °C for 48 h for solvent evaporation. After that, the temperature was slowly increased up to 200 °C in a muffle furnace for 4 h. In the last step, the temperature was increased rapidly, and the membranes were calcinated in the same muffle furnace at 400 °C for 4 h.

2.4. Fabrication of PSF-P123-NP coated nanocomposite ceramic membrane

A porous polymer membrane layer with nanoparticles blended in it was deposited on the tubular ceramic membrane to make the second nanocomposite ceramic membrane. The surface of the tubular ceramic membranes was first scoured with a 400-grit sandpaper to make the surface smooth, and then washed with DI water thoroughly. Polymer blend was prepared by dissolving 17% w/w PSF, 3% w/w P123, and 1% w/w NP in DMAc solvent and stirring the solution overnight, followed by ultrasonication until homogeneity was achieved. After the polymer-NP blend was prepared, the ceramic membranes were dipped in it for 10 s and quickly kept in a refrigerator (-15 °C) for the next 10 s. After that, it was kept in contact with humid air (relative humidity 70–80%) for 1 min to cause vapour-induced phase separation before immersing it in DI water to cause non-solvent-induced phase inversion. After immersion in DI water, the polymer membrane layer was formed on the tubular ceramic membrane. Membranes were dried in ambient air for the next 12 h before using.

2.5. Fabrication of PVA-NP-coated nanocomposite ceramic membrane

The third membrane was coated with a hydrophilic polymer–NP blend using interfacial polymerisation. A solution was prepared by mixing 5% w/w PVA with 50 ml of water and 1% w/w NPs by heating the mixture at 90 °C while being stirred. The PVA and NPs were added gradually while stirring the solution to maintain homogeneity. Once a homogeneous solution was formed, the temperature was turned off, and the solution was cooled down to room temperature to add 2% w/w glutaraldehyde and 0.5% w/w HCL to the homogeneous solution to facilitate crosslinking. After the addition, the solution was stirred for an hour without heat and ultrasonicated for 10 min. The ceramic membrane was dipped in the resulting PVA-NP blend solution, followed by air drying for 24 h. At last, the coated, air-dried ceramic membrane was kept in a muffle furnace at 110 °C for 1 h to finish the membrane fabrication process.

2.6. Characterisation of the adsorbent nanoparticles

Transmission electron microscope (TEM) imaging (JEOL, JEM2100F) was used to determine the nanoscale structure of the adsorbent nanoparticles. X-ray diffraction (XRD) using Ultima III, Rigaku, with a radiation wavelength: 1.540560 Å, was done to determine the structure and crystallinity of the NPs, Brunauer–Emmett–Teller (BET) surface area analyser (N₂ adsorption isotherm at –196 °C using Quantachrome Autosorb iQ) was used to determine the total surface area of the NPs and the nature of its pores, and X-ray photon spectroscopy (XPS) was done using the instrument made by Omicron NanoTechnology GmbH to determine the composition of the NPs. The zeta potential and pH of zero-point charge was measured using Zetasizer by Malvern Panalytical.

2.7. Simultaneous heavy metal adsorption and competition

Adsorption experiments were conducted in batch mode; a 0.5 g/l dose of the adsorbent was used in the case of multi-metal adsorption experiments. A summary of adsorbent properties is given in [Table 1](#). Arsenic and chromium can exist in water as both neutral compounds and anions. Arsenic species, As(III), occur as H₃AsO₃, and H₂AsO₃[–], among others. Chromium is found in the form of Cr(VI) or trivalent Cr(III), appearing as CrO₄^{2–}, HCrO₄[–], etc., while Pb is found in water as Pb²⁺ up until pH ~9, after that Pb(OH)^{3–} becomes the dominant species [\[22\]](#), [\[23\]](#). To check the effect of anions and cations on the competition, experiments were done with 4 metal mixed solutions (As, Pb, Cd, Cr), two cationic and one anionic species (As, Cr, Pb; As, Cr, Cd, and Cd, Cr, Pb) and cation-cation (Pb, Cd), anion-anion (As, Cr), cation-anion (As, Cd) combinations for 1 – 20 mg/l concentration. At the end, the adsorbate was passed through a membrane syringe filter to remove adsorbents, and the resulting liquid was centrifuged at 10,000 rpm. Concentrations of the heavy metals were measured using an inductively coupled plasma optical emission spectrometer (ICP-OES, model: Optima 8000, PerkinElmer).

Table 1. performance comparison of ceramic composite membranes.

Membrane type	Fabrication method	Properties	performance	Ref.
ZnO nanorods or SiO ₂ nanoparticles grafted onto basalt ceramic membrane	Solution deposition method using NaOH, hexamethylenetetramine and autoclave-assisted heating.	Microfiltration <i>E.coli</i> filtration and killing	Highest PWF 841.9 L/m ² h bar 100% <i>E. coli</i> removal from an initial concentration of > 2.4 × 10 ⁶ MPN/L	[14]

Membrane type	Fabrication method	Properties	performance	Ref.
			74.90% solid-surface antimicrobial activity	
ZrO ₂ -SiO ₂ composite ceramic membranes embedded by silver nanoparticles	dip-coating followed by drying and calcination	Dye rejection Anti-bacterial Anti-fouling	96.74% dye rejection and PWF of 331.36 L m ⁻² h ⁻¹ for 50 mg L ⁻¹ reactive black KN-B solution at 2 bar pressure FRR of 91.18% after three filtration cycles antibacterial efficiency of 98.86% against <i>E. coli</i> and 97.42% against <i>S. aureus</i>	[45]
Ceramic supported attapulgite-graphene oxide composite membrane	vacuum-assisted filtration method	Heavy metal removal Enhanced hydrophilicity	maximum PWF of 6.3 L m ⁻² h ⁻¹ and rejection (99.9%) was achieved for Cu ²⁺ , Ni ²⁺ , Pb ²⁺ and Cd ²⁺ Lowest water contact angle found ~38.4° 3500 min continuous Cu ²⁺ rejection	[46]
Chitosan coated kaolin ceramic ultrafiltration membrane	crosslinked chitosan over the top layer of ceramic support using dip-coating	Hg(II) and As(III) removal Low-cost	100% removal for both 500 µg L ⁻¹ Hg(II) and 1000 µg L ⁻¹ As(III) Stable flux 3.44 × 10 ⁻⁶ m ³ m ⁻² s ⁻¹ for 103.4 kPa transmembrane pressure	[47]
EDTA modified ceramic-polyethyleneimine composite membrane	Vacuum filtration assisted deposition of EDTA	As(V), Cr(VI) and Cu(II) removal	Pure water permeability 8 L m ⁻² h ⁻¹ bar ⁻¹ , lowest contact angle 49.1° ± 1° Highest rejections: 99.82% for Cu(II), 96.75% for As(V) and 97.22% for Cr(VI)	[20]

Membrane type	Fabrication method	Properties	performance	Ref.
Mg-Al-Ti oxide incorporated ceramic-polymer nanocomposite membrane	1. In-situ NP grown	As, Cr, Cd, Pb	> 90% removal in 3–10 pH range	This work
	2. PVA-NP coated	removal	98.74% solid-surface antimicrobial activity & 83.52% biofilm inhibition	
	3. PSF-P123-NP coated	Anti-fouling	95.47% flux recovery	
		Anti-microbial	Lowest water contact angle found	
		Enhanced hydrophilicity	~25.06°	

2.8. Characterisation of the membranes

Membrane morphology and the nature of the coatings were probed by an FESEM (model: Inspect F50, FEI). EDAX mapping was done to visualise the incorporation of the NPs in the coating and the membrane matrix in the case of an in-situ NP grown membrane. Traces of arsenic after filtration were found through the EDAX mapping of the used membrane. The contact angles of the MMMs were measured using a goniometer following the sessile drop method at room temperature (model: Apex instruments, ACAMNSC 30). Equilibrium water content (EWC) or water absorption and apparent porosity of the membranes were determined using the following equations:

$$EWC\% = \frac{W_w - W_d}{W_d} \times 100 \quad (1)$$

$$Apparent\ Porosity = \frac{W_w - W_d}{V} \times 100 \quad (2)$$

Here, V is the membrane volume, and the weight of the wet membrane and the weight of the dry membrane are represented by W_w and W_d , respectively. The Experiments were done following the guidelines of ASTM C373–14, with the assumption that 1 cm³ of water corresponds to a mass of 1 g [24].

2.9. Membrane permeation experiments

A dynamic, continuous crossflow filtration setup was used to perform permeation experiments on the fabricated membranes. Newly fabricated tubular membranes of 5 mm internal diameter and 10 cm length were used in the filtration cell and fitted with

3 nuts and bolts. A pump was used to provide transmembrane pressure (TMP). Adjustment of TMP was done by a valve placed in the retentate line (schematic shown in Fig. 7).

2.9.1. Pure water flux of the membranes (PWF)

PWF of the nanocomposite membranes were measured by permeating DI water through the membranes at different trans-membrane pressure points in the crossflow setup. PWF (J_w) was measured by the equation:

$$J_w = \frac{Q}{A \Delta t} \quad (3)$$

Where 'Q' is the volume of permeate in Litres, 'A' is the effective filtration area in m², and ' Δt ' is the permeation time in hours. The slope of J_w vs Pressure plot was defined as hydraulic permeability.

2.9.2. Anti-fouling property

The anti-fouling property of the membranes was studied by fouling the membrane with a 1000 mg/l aqueous solution of BSA until the equilibrium BSA flux was measured (J_p). After the BSA run, the membranes were manually washed with DI water and subjected to DI water permeation again to determine the PWF after fouling (J_{w2}).

$$FRR = \frac{J_w}{J_{w2}} \times 100\% \quad (4)$$

$$R_t = \left(1 - \frac{J_p}{J_w}\right) \times 100\% \quad (5)$$

$$R_r = \left(\frac{J_{w2} - J_p}{J_w}\right) \times 100\% \quad (6)$$

$$R_{ir} = R_t - R_r \quad (7)$$

Eq. 7 to 10 were used to determine flux recovery ratio (FRR), Total fouling ratio (R_t), reversible fouling ratio (R_r), and irreversible fouling ratio (R_{ir}).

2.10. Membrane heavy metal removal performance

Experiments for heavy metal removal were run in both continuous-crossflow mode, where the heavy metal-laden feed was continuously given to the feed tank and only the retentate stream was recycled, or total recycle mode, where permeate was recycled along with the retentate stream to the feed chamber. For long-term filtration experiments, intermittent collection of samples from the permeate stream was done; for other experiments, the concentration of the accumulated permeate was taken to calculate rejection.

2.10.1. Single metal removal: long-term performance and pH dependence

Single metal solutions of 1000 µg/l concentration in the pH range of 3–13 were used to determine the pH dependence of the filtration performance for all three membranes. 0.1 N NaOH and 0.1 N HCl solutions were used to change the pH. Single metal long-term filtration experiments were conducted using 1000 µg/l solutions of As(III), Pb(II), Cr(VI) and Cd(II) to ascertain the stability and breakthrough volume of the prepared membranes. The experiment was terminated when the arsenic concentration in the permeate exceeded the WHO-suggested allowable limit for the metal concerned. According to WHO and BIS, As(III) and Pb(II) have a suggested minimum safe concentration of 10 µg/l, while Cr(VI) and Cd(II) have the minimum safe concentration limit of 50 µg/l and 3 µg/l, respectively.

2.10.2. multi-metal: simulated and real samples

To study the simultaneous removal of heavy metals, at first, a simulated solution, a mixture of 4 heavy metal aqueous solutions, each with a 250 µg/l concentration, was used. The experiment was conducted by filtering 10 litres of the solution. Real heavy metal contaminated surface water samples were collected from a pond in East Kolkata Wetlands (22° 0' 27" N 88° 0' 27" E). The water was analysed using ICP-OES to find concentrations of the heavy metals in it. Primarily, 650 µg/l As, 812.4 µg/l Pb, 1956 µg/l Cr and 343 µg/l Cd were found along with trace amounts of Mn, Fe, Ni and Cu. The contaminated water was filtered through the best-performing membrane from previous experiments to study the breakthrough performance and the multi-metal removal efficiency of the membrane in practical applications.

2.11. Membrane anti-microbial performance

In our previous work, Mg-Al-Ti oxide NPs showed antimicrobial properties against both gram-positive and gram-negative bacteria. Here, the anti-microbial studies were done to evaluate the extent of the anti-microbial activity of different

nanocomposite membranes to ultimately assess the efficiency of different methods of nanoparticle incorporation into the ceramic membranes.

2.11.1. Disk diffusion assay

The antimicrobial efficacy of the nanocomposite membranes was assessed using the disk diffusion method. Briefly, a suspension of bacteria was prepared, inoculating *E. coli* in Luria Bertani broth media and incubating at 37°C, 150 rpm for 24 h and was adjusted to a cell concentration of 1×10^8 CFU/ml. Muller Hinton agar plates were prepared, 0.1 ml of *E. coli* suspension was spread onto the agar plate and rested for 10 mins. Nanocomposite membranes (cut into ring shape, width: 5 mm) were aseptically placed and gently pressed onto the inoculated agar surface to ensure proper contact. A sterile, pristine membrane was used as a control. The plate was incubated for 24 h at 37°C under aerobic conditions. Following incubation, the antimicrobial activity was determined by measuring the diameter of the clear zone surrounding the membrane, referred to as the zone of inhibition (in mm). The experiment was performed in triplicate and under a sterile environment.

2.11.2. Solid surface antimicrobial activity

A solid surface antibacterial assay was conducted, modifying a previously reported process [25] to evaluate the antibacterial performance of the nanocomposite membrane on direct contact with test bacteria. Tubular sections of sterile nanocomposite membranes were cut (length: 2 cm), and 15 µl *E. coli* suspension containing a standardised cell concentration of 1×10^8 CFU/ml was inoculated directly on the surface of the membrane and was placed on the petri-plate, followed by overnight incubation at 37°C under aerobic conditions. After the incubation period, the nanocomposite membranes were dipped into a 0.9% sterilised NaCl solution and kept in an incubator shaker for an hour at 150 rpm to recover the surviving bacteria attached to the surface of the membranes. Further, serial dilutions were prepared and plated on nutrient agar to determine the number of viable colonies (CFU/ml). Unmodified membranes served as a control, and the solid surface antimicrobial activity of the nanocomposite membrane was calculated by comparing the number of viable bacteria recovered from treated samples with those recovered from control surfaces. It was expressed as a logarithmic reduction in bacterial count, providing a quantitative measure of surface-associated antibacterial activity.

2.11.3. Anti-biofilm assay

The antibiofilm activity of the nanocomposite membranes was qualitatively and quantitatively assessed using the crystal violet (CV) staining method, a widely employed technique for evaluating biofilm biomass formation. The assay was performed against biofilm-producing bacteria *Staphylococcus aureus* (NCIM-2679). Briefly, *S. aureus* was grown in tryptic soy broth with 0.1% glucose supplement for 24 h at 37°C and shaking at 150 rpm. The bacterial culture was diluted to obtain a cell density of 1×10^8 CFU/ml. The nanocomposite membranes were cut into a length of 2 cm and incubated with freshly inoculated media at 37°C for 3 days under static conditions to allow the formation of biofilm. After 3 days of incubation, the spent media were decanted, and the nanocomposite membranes were washed with PBS (phosphate buffer saline - pH 7.4) buffer to remove the planktonic cells. Subsequently, 100 µl of 0.1% (w/v) crystal violet solution was added to the nanocomposite membranes and incubated for 30 mins, followed by rinsing with PBS and distilled water to remove the unbound stain. Furthermore, to solubilise the bound crystal violet, 200 µl of 95% ethanol was added to the nanocomposite membranes and incubated for 15 mins. Qualitative analysis was done by observing the nanocomposite membranes for retention of the stain on the surface, and quantitative determination was done by measuring the absorbance of the resulting solution at 595 nm [26]. The overall biomass of the biofilm is correlated with the absorbance intensity. A pristine ceramic membrane was used as a control. The percentage anti-biofilm inhibition was calculated using the formula:

$$\text{Percentage Inhibition (\%)} = \frac{Abs_{Control} - Abs_{Treated}}{Abs_{Control}} \times 100\% \quad (11)$$

where $Abs_{control}$ is the absorbance for pristine membranes, $Abs_{treated}$ is the absorbance for nanocomposite membranes. A significant reduction in OD595 values indicates effective inhibition or disruption of biofilm formation by the nanocomposite membranes. The experiment was performed in triplicate in a sterile environment to ensure reproducibility.

2.11.4. E. coli filtration experiment

The bacterial filtration performance of the fabricated nanocomposite membranes was carried out using *E. coli* as the model organism. The assay was intended to evaluate the membrane's capacity to physically retain or inactivate bacterial cells during the filtration. Precisely, 50 ml of nutrient broth media was taken and was inoculated with *E. coli*, incubated for 24 h at 37°C in an incubator shaker with 130 rpm agitation. The bacterial suspension was subsequently diluted with sterile phosphate-buffered saline (PBS, pH 7.4) to obtain a working cell concentration of approximately 1×10^8 CFU/ml, as confirmed by optical density measurements and serial dilution plating. 100 ml of the working bacterial solution was passed through the nanocomposite membranes over 60 min at a fixed pressure. Permeates were collected at 10-minute intervals, and the number of viable bacteria

in the permeates was quantified by serial dilution and plating onto nutrient agar, followed by incubation at 37°C for 24 h. Colony-forming units (CFU) were enumerated to determine the bacterial removal efficiency (BRE) of the nanocomposite membranes. An unmodified pristine membrane was used as a control in this study to evaluate the efficiency of the nanocomposite modified membrane.

2.12. Membrane stability: leaching, desorption, and regeneration

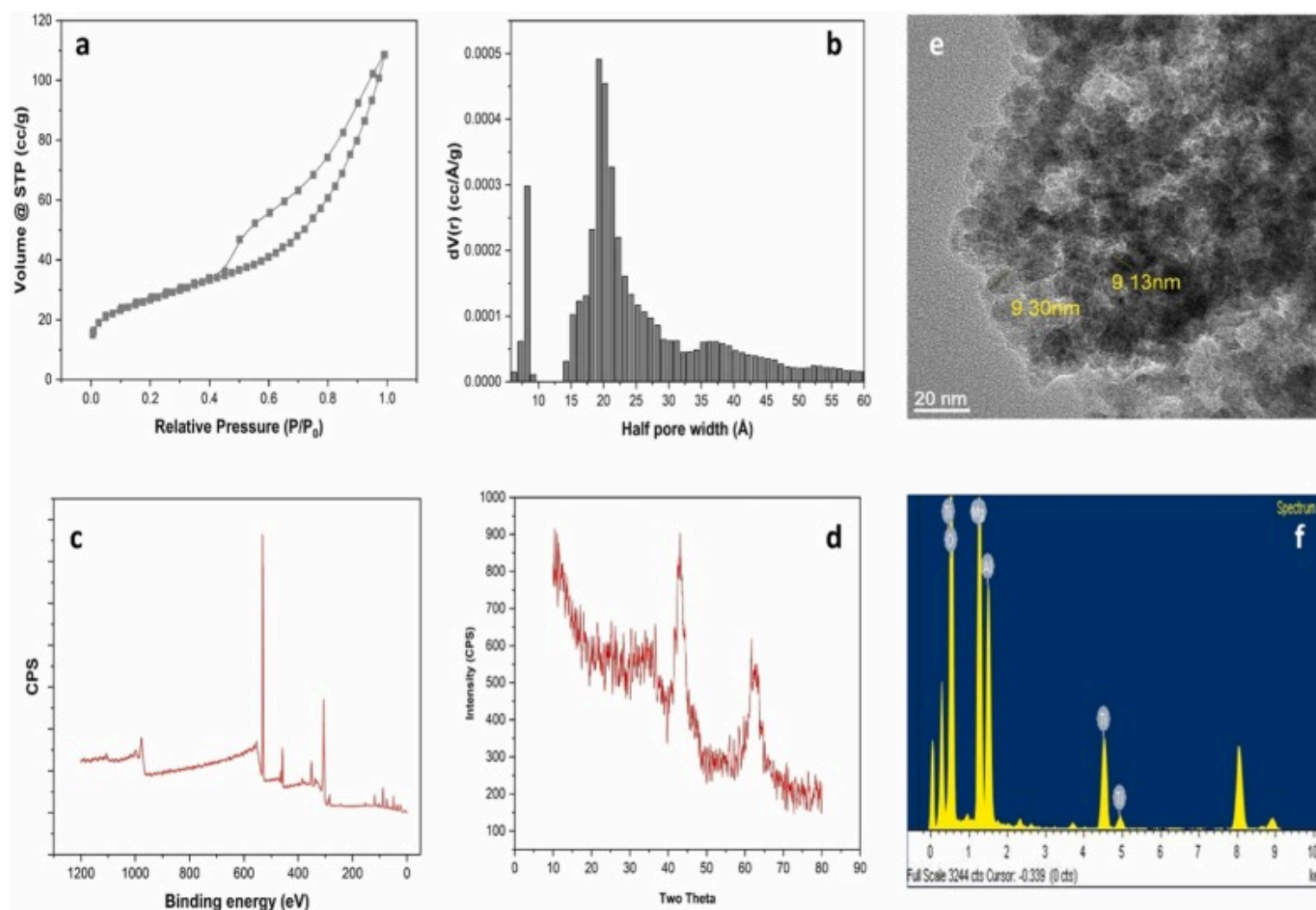
The potential leaching of adsorbents from nanocomposite membranes poses a challenge to their long-term stability and practical application. To assess this, the concentrations of magnesium (Mg), aluminium (Al), and titanium (Ti) in the permeate were monitored during the filtration of an arsenic solution containing 1000 µg/l As(V) in a wide pH range (3–13). Desorption was checked by storing the membrane in water at neutral, acidic, and basic pH (5–10). ICP-OES was used for determining the metal concentrations. Membrane regeneration and flux recovery were carried out using a strong basic solution of 1 N NaOH, followed by 1 N HCl to neutralise the embedded adsorbents. To evaluate the membranes' performance post-regeneration, two long-term filtration runs using 1000 µg/l As(III) as the contaminant were conducted by repeating the regeneration process.

3. Results and discussion

3.1. Characterisation of the nanoparticles

Morphological, structural, and chemical properties of the nanoparticle adsorbents were studied using XRD, TEM, EDAX and BET surface area analysis. Fig. 1 contains a compilation of the results of those characterisation studies. In Fig. 1a, the N₂ adsorption isotherm of the NP is a type-IV isotherm with H2-type hysteresis loop, suggesting the presence of mesopores. Fig. 1b shows the pore size distribution, where the half-pore size determined by the non-local density functional theory (NLDFT) method was 1.96 nm. The XPS spectra of the NP are shown in Fig. 1c. The XPS long scan confirms the presence of peaks for Mg 2p (49.5 eV), Mg 2s (87.85 eV), Al 2p_{3/2} (72.5 eV), Al 2s (118.8 eV), Ti 3p (35.5), Ti 2p_{3/2} (458.08 eV), Ti 2p_{1/2} (463.46 eV). XRD pattern of the NP, shown in Fig. 1d, confirms the presence of the cubic phase of MgO with peaks for (002) and (202) planes at 43°, 62° respectively. No clear peaks are visible for Al and Ti as the material may be amorphous humps in low 2θ [27]. UHR-FEG-TEM (ultra-high resolution field emission gun transmission electron microscope) image shown in Fig. 1e suggests that the average size of the NPs is around 9–10 nm, and the NPs form a honeycomb-like network. The EDAX spectra in Fig. 1f also confirm the presence of Mg, Al

and Ti in the nanoparticles prepared. The pH of zero-point charge for the NPs was found to be 13.05. This high pH_{zpc} may explain electrostatic interactions with the heavy metal ions.



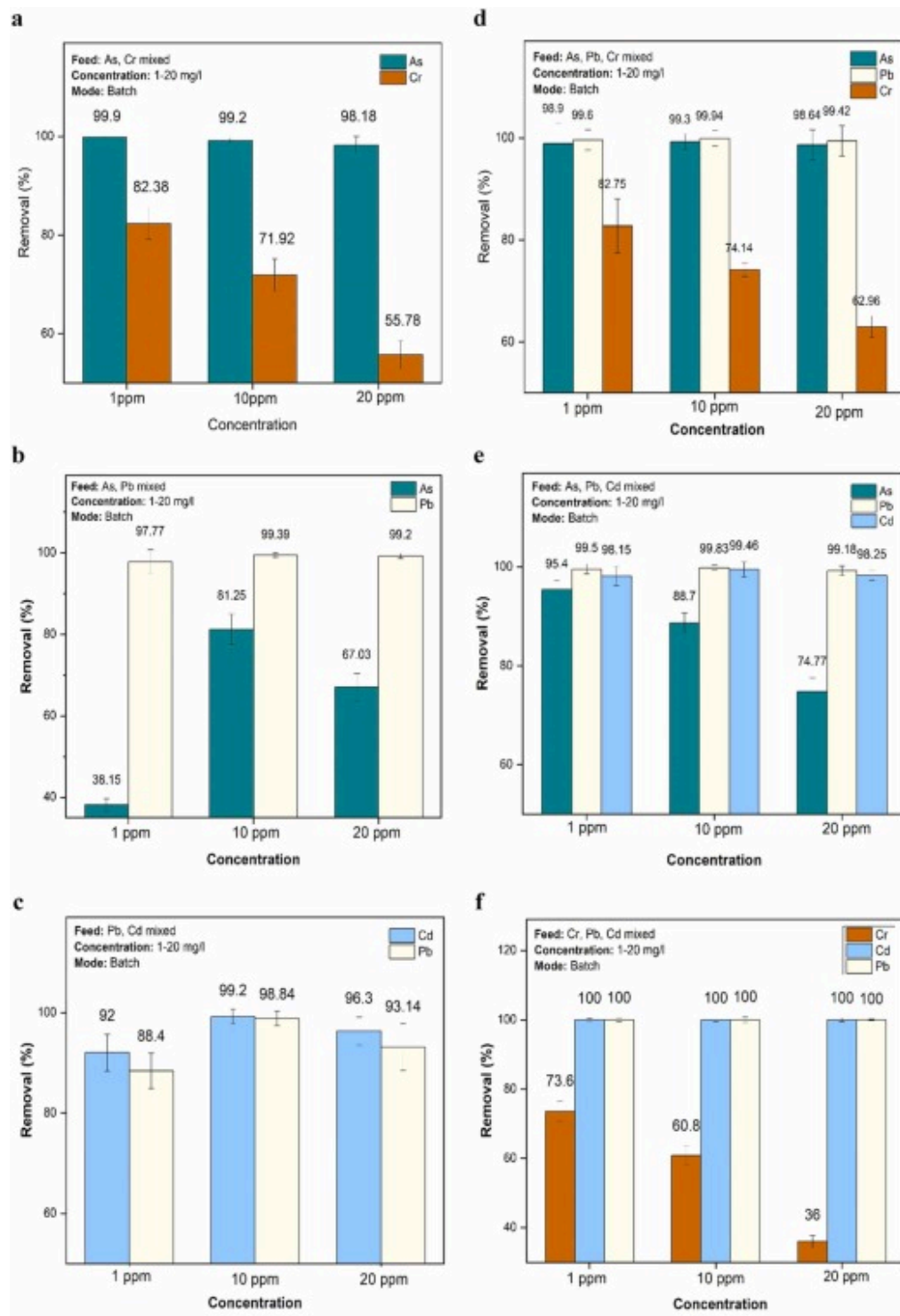
[Download: Download high-res image \(419KB\)](#)

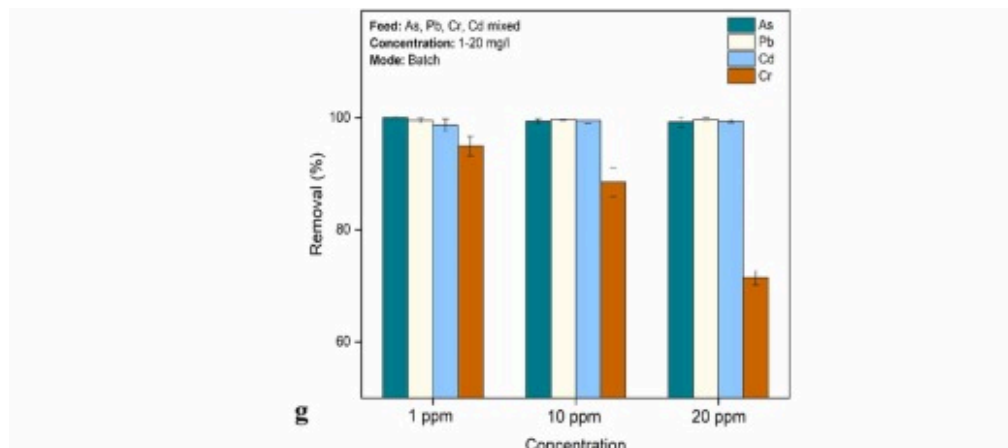
[Download: Download full-size image](#)

Fig. 1. a) N₂ adsorption isotherm, b) pore size distribution (NLDFT method), c) XPS spectra long scan, d) XRD spectra, e) TEM image f) TEM-EDAX spectra of the Mg-Al-Ti oxide NPs.

3.2. Simultaneous heavy metal adsorption and competition

Investigation of simultaneous multi-metal and competition was studied using binary, ternary and quaternary mixtures of As, Pb, Cd and Cr at different concentrations. Since the adsorbent can adsorb both anionic and cationic species [12]. The mixtures were made keeping that in consideration. At pH~7, As(V) exists as H_2AsO_4^- and HAsO_4^{2-} ; Pb exists as majorly Pb^{2+} ; Cr(VI) exists as HCrO_4^- ; and Cd(II) exists as Cd^{2+} . On this basis, we have created anion-anion mixture As-Cr, cation-cation mixture Pb-Cd, cation-anion mixture As-Pb, tri-metallic solutions with anion-anion-cation, anion-cation-cation compositions have also been studied. Mg-Al-Ti oxide shows a pH of zero-point charge at ~13. Due to this high pH_{ZPC} the surface has more positive charges than negative charges at pH~7, which should favour adsorption of anionic species through electrostatic attraction. But, besides electrostatic interactions, the adsorption of heavy metals is suggested to be happening by abundant -OH groups that participate in inner or outer sphere complex formation. The abundant -OH groups can receive and donate H^+ creating $-\text{O}^-$ or $-\text{OH}_2^+$, which allows it to form complexes with both anionic and cationic species [19]. Fig. 2 shows the bar graphs denoting the removal percentage of different heavy metals from the mixture. Fig. 2 g shows that the adsorbent has a lower affinity toward Cr(VI), while it can achieve up to 99% removal for As, Pb, and Cd in a quaternary mixture. In the 1–20 mg/l concentration range, the removal percentages of As, Pb, and Cd are comparable, and no clear selectivity was observed. In the case of the ternary anion-anion-cation mixture (As, Cr, Pb), the removal percentage for Pb is slightly higher than As, while Cr(VI) is still the least adsorbed species. For the cation-cation-anion mixture (As, Pb, Cd), a clear trend is visible. It may be possible that in the presence of cationic species, the surface becomes more negatively charged. This reduces the adsorption of anions created by As. Moreover, due to the relatively smaller size of Pb^{2+} and Cd^{2+} they get higher mobility and more access to the pores, which is why in the binary mixture Pb-Cd, Cd adsorption is favoured as Cd^{2+} has even less size than Pb^{2+} [28]. For the anion-anion mixture of As and Cr, clear selectivity for As is visible. The reason for less Cr adsorption may be due to the very high hydrated radius of HCrO_4^- ions, which (0.5–1 nm) is less favourable to the adsorbent pores than smaller-sized HAsO_4^{2-} ions [29]. Finally, for the cation-anion mixture of As and Pb, the adsorbent is more selective towards Pb, confirming the result from the ternary mixture adsorption.





Download: [Download high-res image \(454KB\)](#)

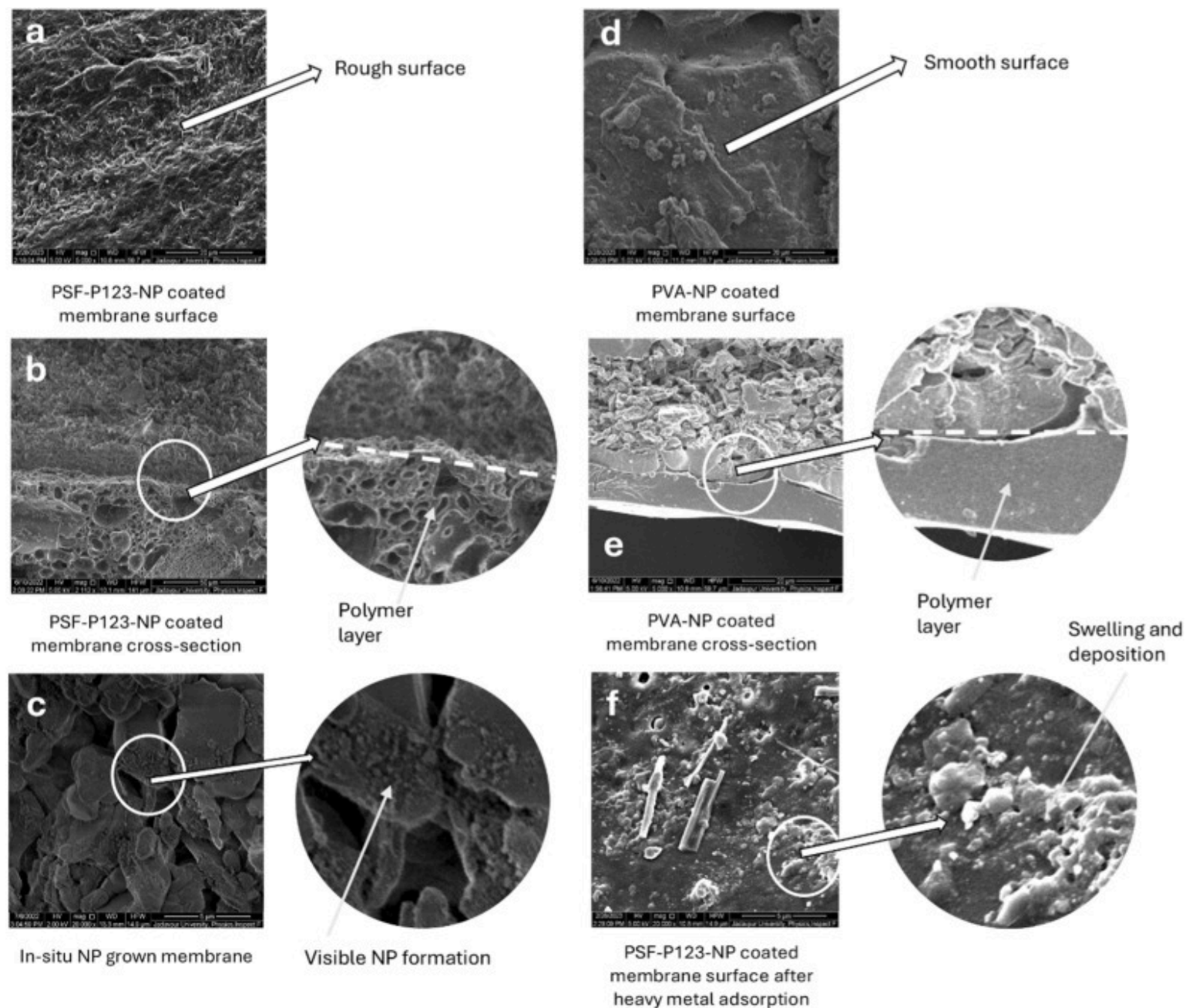
Download: [Download full-size image](#)

Fig. 2. a,b,c: binary competition, d,e,f: ternary competition, g: quaternary competition.

3.3. Characterisation of the membranes

Lateral cross-sections and the surfaces of the 3 novel membranes were probed using FESEM imaging to visualise and confirm the nature of the coatings and nanoparticle addition. Fig. 3 represents the FESEM images of PVA-NP-coated, PSF-P123-NP-coated, and in-situ NP grown ceramic nanocomposite membranes. A cross-section of the in-situ NP grown membrane was not probed since it does not have any coating. In Fig. 1b, the PSF-P123-NP coating layer is visible with sponge-like structured pores. It may be taking

this sponge-like structure due to the passing of humid air after taking it out of the refrigerator [30]. By controlling the humidity of the passing air, the structure may also be tuned. No visible gap between the membrane and the coated layer confirms the success of the coating process. Fig. 1d shows the cross-section of the PVA-NP coated membrane; the coated layer is clearly visible and shows a less porous nature than the previous one. Fig. 1e shows the magnification of the surface of the in-situ NP grown membrane, where deposition of NPs is clearly visible.

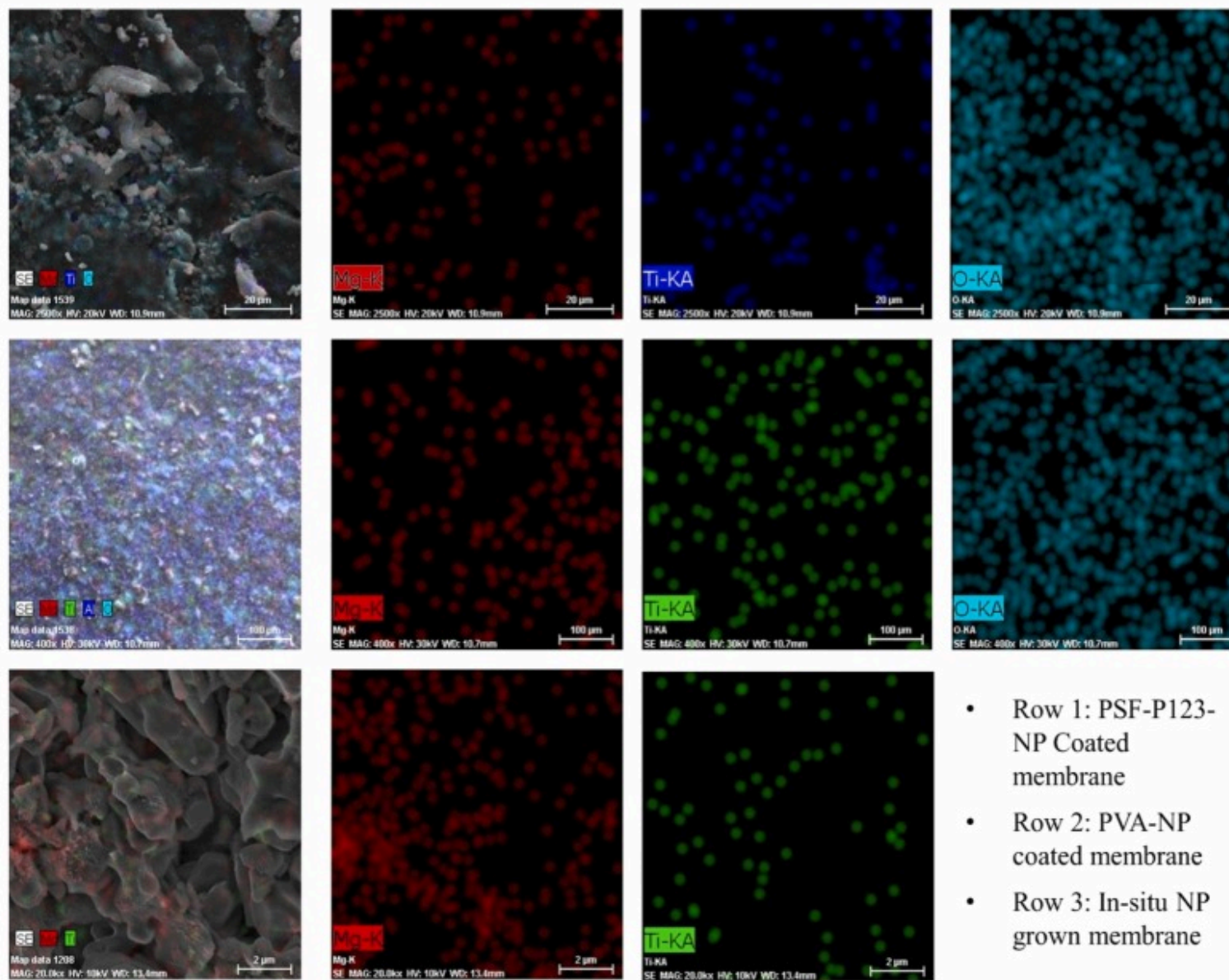


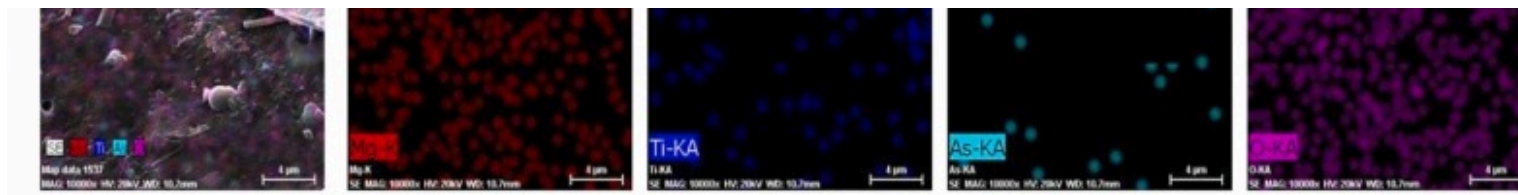
[Download: Download high-res image \(785KB\)](#)

[Download: Download full-size image](#)

Fig. 3. a) PSF-P123-NP coated surface, b) PSF-P123-NP coated membrane cross-section, c) in-situ NP grown membrane surface, d) PVA-NP coated membrane surface, e) PVA-NP coated membrane cross-section, f) Surface of PSF-P123-NP coated membrane after heavy metal adsorption.

Fig. 3f shows the surface of the PSF-P123-NP membrane after long-term heavy metal filtration. It is visible that the roughness of the surface has decreased, and some swelling has occurred. Incorporation of the NPs was probed using EDAX mapping of the surface and cross-sections of the membranes. Fig. 4a shows the compilation of the EDAX maps of unused membranes, where the maps for Ti, Mg and O prove the presence of the NPs. However, maps for Al are omitted as the base membrane is made of alumina. Fig. 4b shows the EDAX map of the PSF-P123-NP coated membrane after arsenic filtration, where the map for arsenic shows the confirmation of adsorptive removal.

a**b**

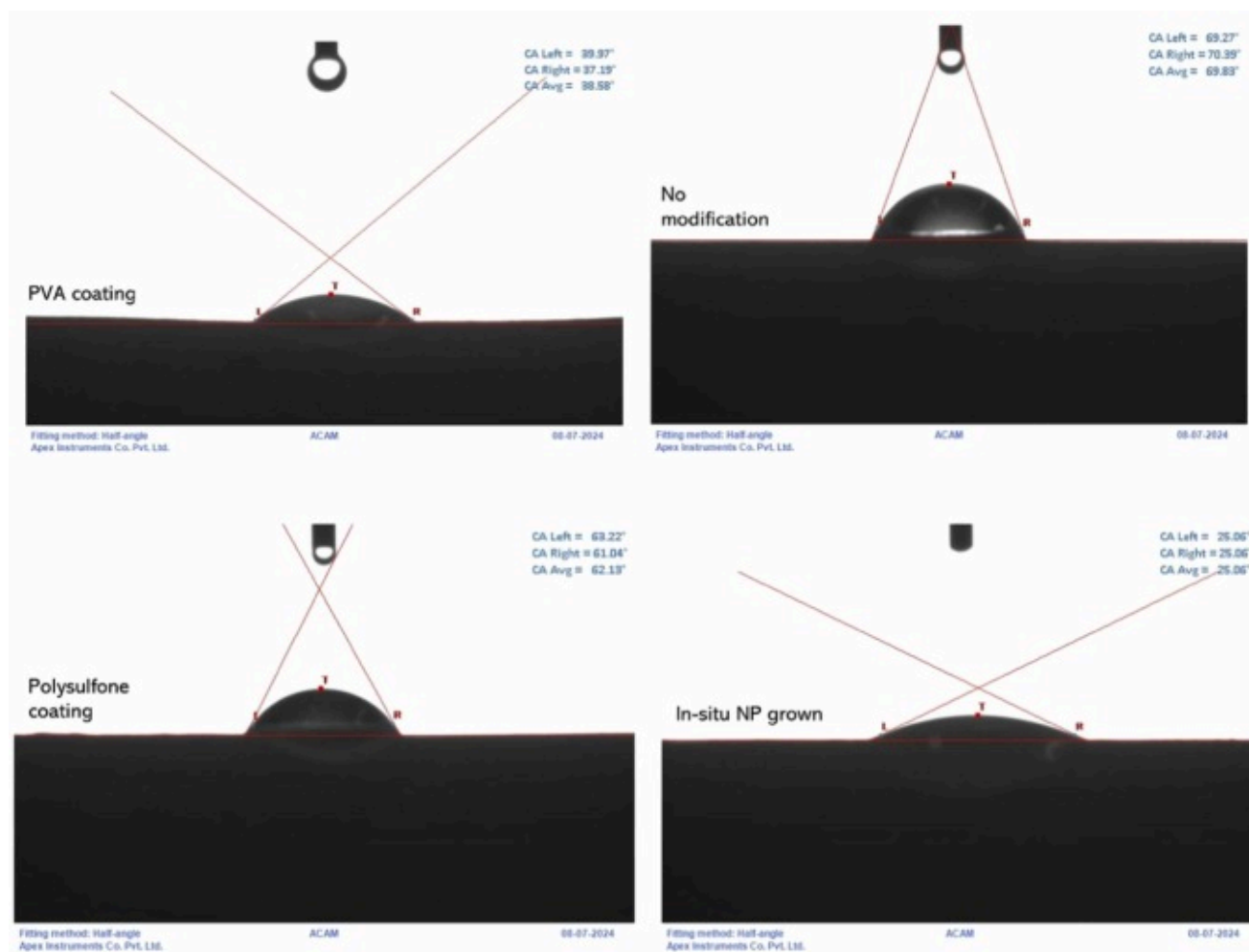


[Download: Download high-res image \(841KB\)](#)

[Download: Download full-size image](#)

Fig. 4. a) EDAX maps of all 3 membranes, b) EDAX map of PSF-P123-NP coated membrane after arsenic adsorption.

Water contact angles for the nanocomposite membranes, as well as a pristine ceramic membrane, were measured to study the hydrophilicity improvement by different methods of nanoparticle incorporation. Fig. 5 shows the contact angle images for all the membranes. The contact angle decreased dramatically from 69.83° for the pristine membrane to 25.06° for the in-situ NP grown membrane. PVA-NP coated membrane also showed a dramatic increase in hydrophilicity with a contact angle of 38.58° while the PSF-P123-NP coated membrane showed a contact angle of 62.13° which suggests that, although polysulfone is a hydrophobic material, the addition of NPs is increasing the hydrophilicity for all the membranes.



[Download: Download high-res image \(153KB\)](#)

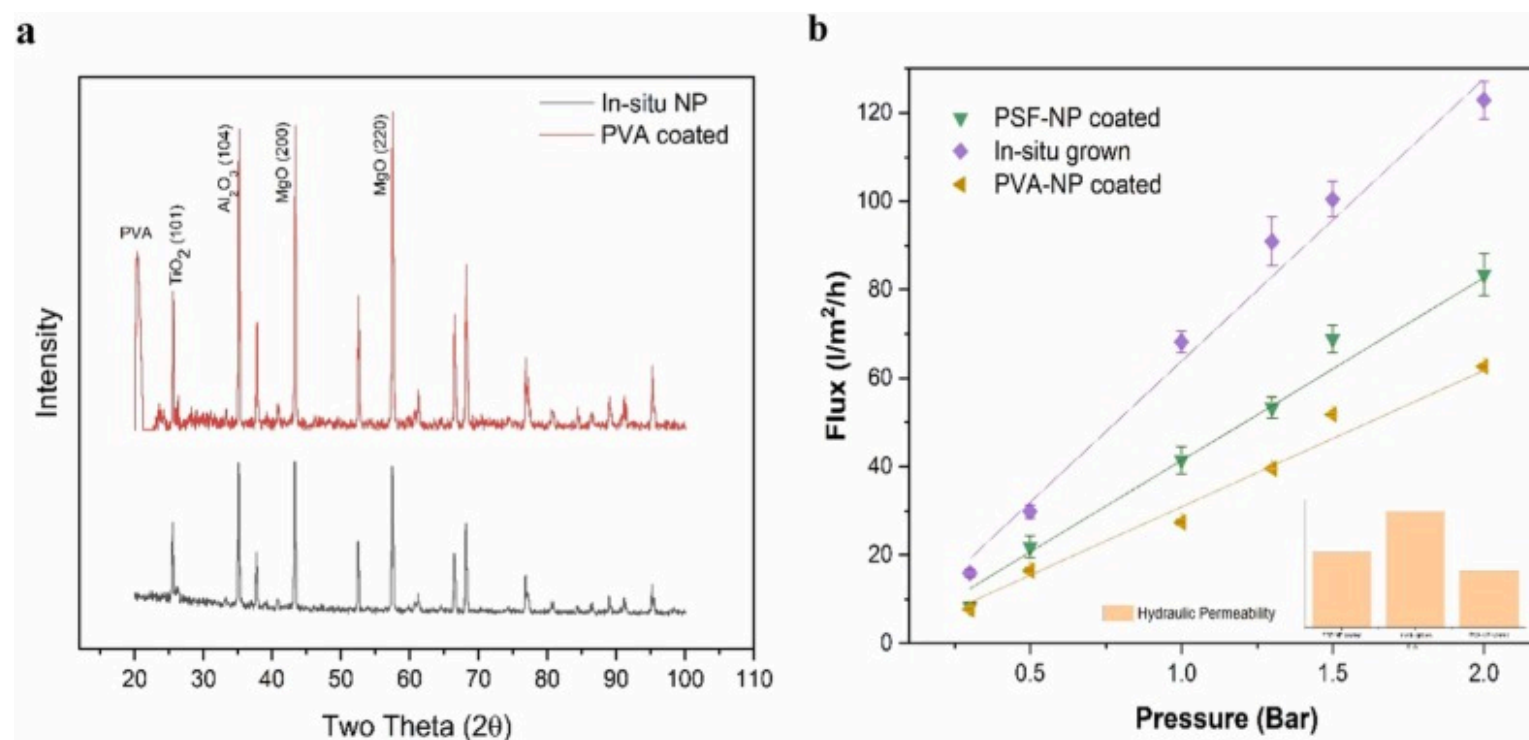
[Download: Download full-size image](#)

Fig. 5. Contact angle measurement for pristine and nanocomposite membranes.

A huge change in EWC was observed after NP incorporation in all three methods. EWC for the pristine alumina membrane was found to be 4.35% while it was increased to 39.65% for the in-situ NP grown membrane, 35.43% for the PVA-NP coated membrane and 32.15% for the PSF-P123-NP coated membrane. This huge increase in EWC can be attributed to the increase in hydrophilicity due to the incorporation of the NPs. Mg-Al-Ti oxide NPs are lowering the contact angle, making the material or the coating more

hydrophilic; hence, the in-situ NP grown membrane is showing the highest EWC. Similar conclusions have been found in the literature [31]. PVA-NP-coated membrane shows higher EWC than the PSF-P124-NP-coated membrane due to PVA being a hydrophilic and water-soluble polymer, which is typically reported to have high water content [32], [33]. Apparent porosity was calculated using Eq. 2, and it was found to be highest for the in-situ NP grown membrane (38.5%), while it is the lowest for the PVA-NP coated membrane (33.5%).

XRD was performed on PVA-NP-coated and in-situ NP grown membranes to confirm NP incorporation. Fig. 6 contains the spectra for the two membranes. In the XRD spectra, different peaks for MgO, TiO₂ and Al₂O₃ can be seen, confirming the incorporation of the NPs, where the sharp peaks correlate to the crystallinity of the NPs.



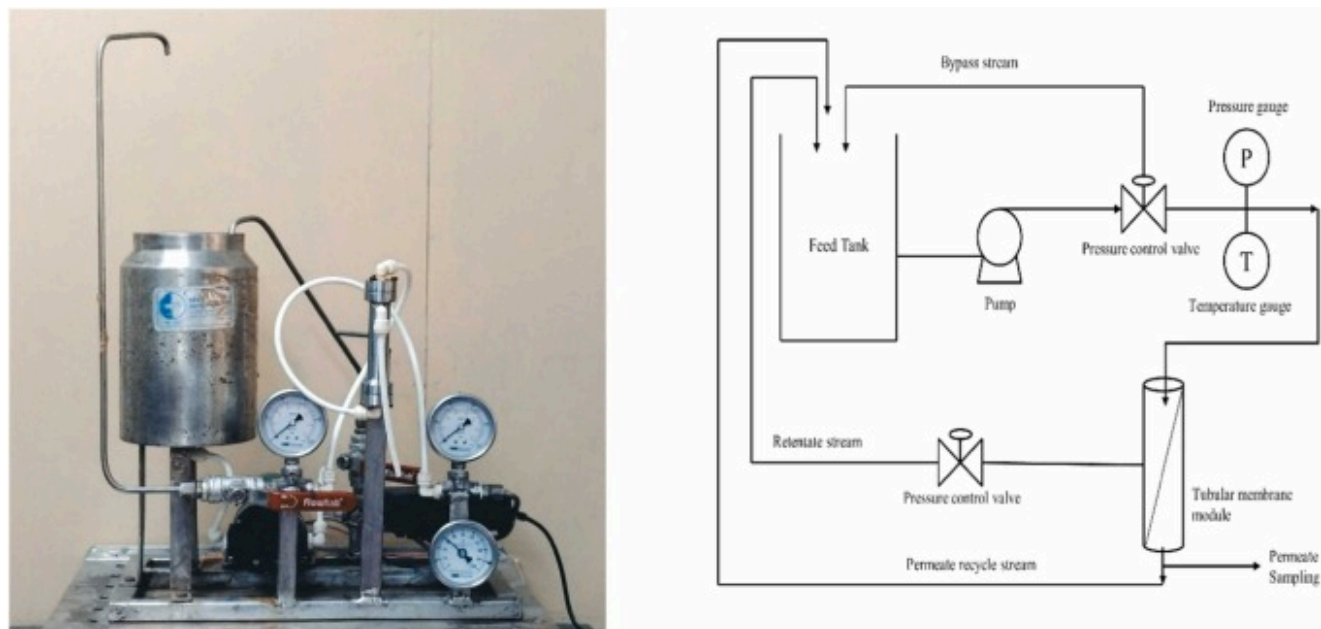
[Download: Download high-res image \(207KB\)](#)

[Download: Download full-size image](#)

Fig. 6. a) XRD spectra for in-situ NP grown and PVA-NP coated membrane, b) Pure water flux and hydraulic permeability of the modified membranes.

3.4. Membrane permeation experiments

Pure water flux, hydraulic permeability and anti-fouling properties of the membranes were studied through different membrane permeation experiments. Fig. 7 shows the schematic and image of the crossflow setup on which the experiments were conducted. PWF of the in-situ NP grown membrane was found to be higher than the other two coating-based membranes, which can be mainly attributed to the decrease in apparent porosity due to the addition of different layers. Besides that, the in-situ NP grown membrane has shown a very low water contact angle, which also supports the result [31]. Although the PWF of the in-situ NP grown membrane is lower than that of the pristine membrane due to pore blockage caused by the growth of NPs.



[Download: Download high-res image \(185KB\)](#)

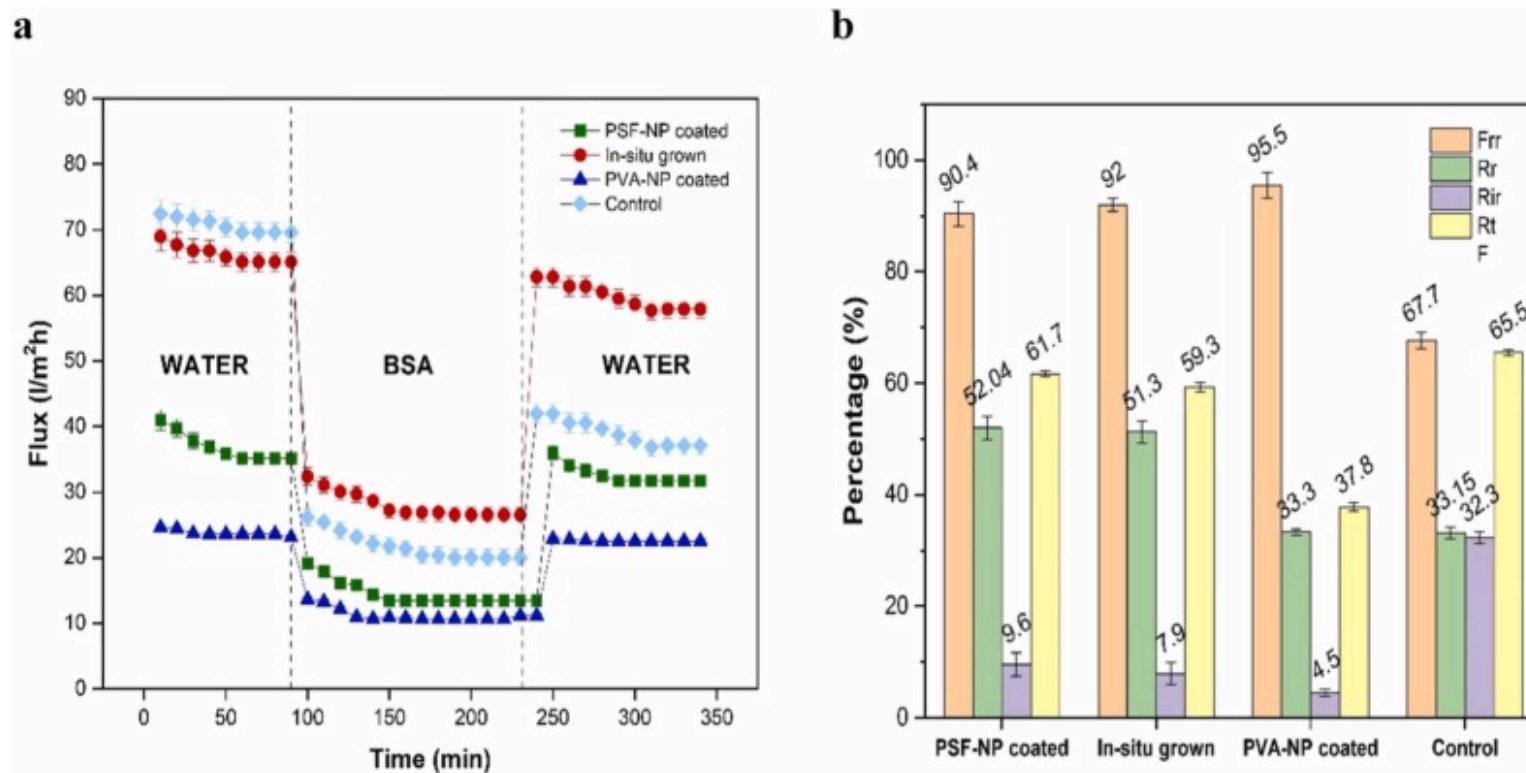
[Download: Download full-size image](#)

Fig. 7. image of the crossflow setup (left), Schematic of the crossflow setup (right).

However, the PSF-P123-NP coated membrane showed a higher PWF than the PVA-NP coated membrane despite having a higher contact angle. The reason for that may be a higher apparent porosity of the coating. Since the PSF-P123-NP coating is done by phase inversion, the sponge-like morphology of the coating layer allows a more porous structure, facilitating a higher flux, while

the PVA-NP coating layer is rather dense, resulting in a relatively lower PWF. Fig. 6b shows the graph for PWF along with a bar chart showing the hydraulic permeability of the membranes, which is the slope of the PWF lines.

The anti-fouling property of the membranes was examined using a 1000 mg/l BSA solution as the foulant. Fig. 8a presents two cycles of PWF-BSA run to assess flux recovery ratio and other parameters, as discussed in Section 2.9.2. The study revealed that the pristine membrane achieved a 67.66% flux recovery after fouling and physical cleaning in the second run, while the PVA-NP coated membrane showed the highest flux recovery of 95.47%. The in-situ NP grown membrane and the PSF-P123-NP-coated membrane exhibited flux recoveries of 92% and 90.4%, respectively. It can be inferred that the inclusion of NPs enhances the anti-fouling properties of the modified membranes. A hydrated layer formed by hydrogen bonding and increased hydrophilicity, facilitated by abundant -OH groups on the NPs, likely contributes to this anti-fouling effect. This layer reduces the accumulation and sorption of BSA [31]. Furthermore, BSA has an isoelectric point at $\text{pH} \sim 4.6\text{--}5.5$, and a 1000 ppm BSA solution shows a pH around 7, which means the BSA molecules carry a net positive charge. Since the NPs have a pHzPC of ~ 13 , the surface of the membrane will also be positively charged at $\text{pH} \sim 7$, creating electrostatic repulsion that reduces the irreversible fouling. The surface roughness often correlates with the hydrophilic/phobic nature of the surface. For a surface that is already hydrophilic, increasing roughness usually lowers the water contact angle and strengthens hydrophilicity; for an already hydrophobic surface, roughness usually raises the contact angle and strengthens hydrophobicity [34].



[Download: Download high-res image \(264KB\)](#)

[Download: Download full-size image](#)

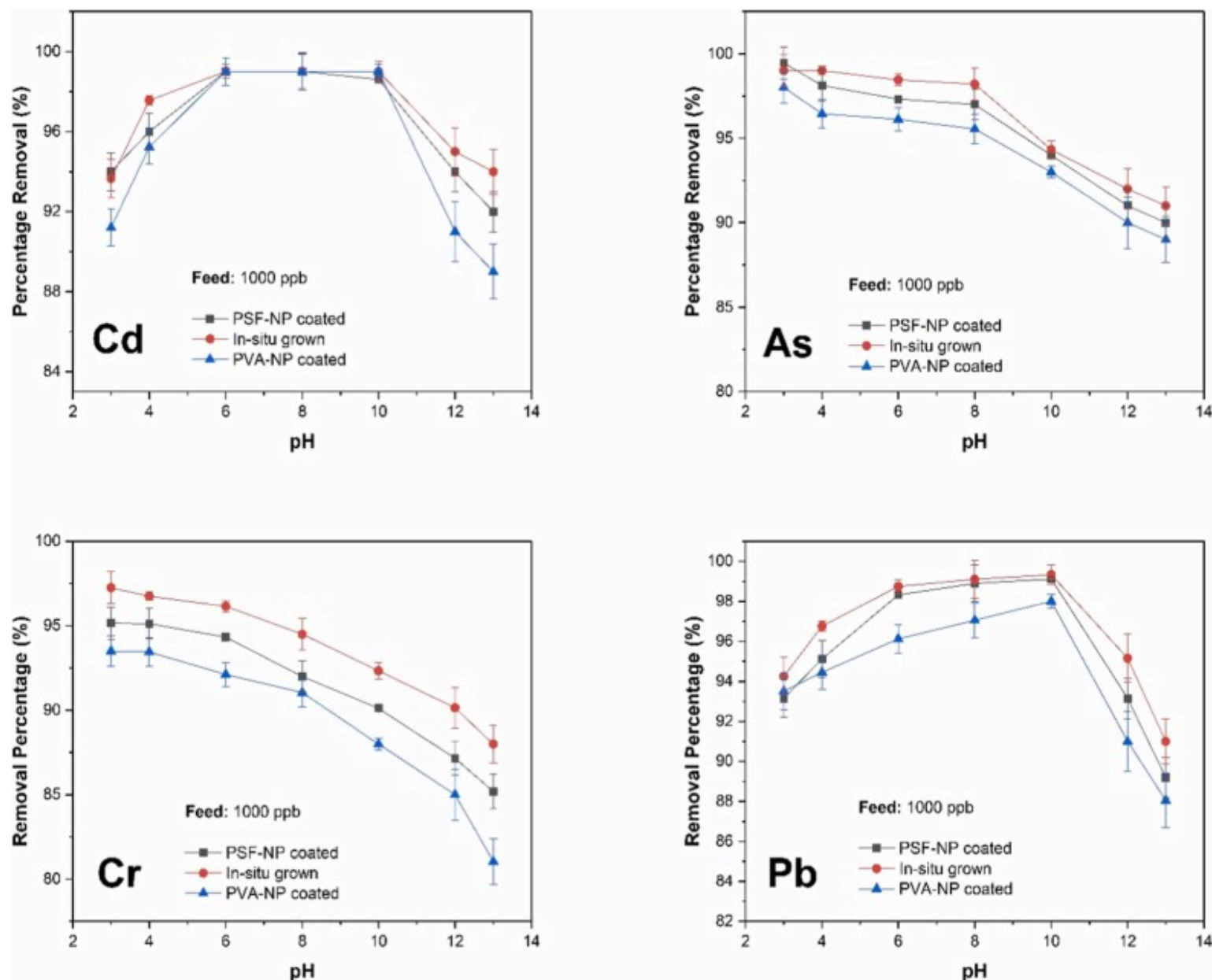
Fig. 8. a) PWF-BSA-PWF permeation experiment for all the membranes, b) Flux recovery and fouling parameters for all the membranes.

This layer reduces the accumulation and sorption of BSA [31]. The surface roughness often correlates with the hydrophilic/phobic nature of the surface. For a surface that is already hydrophilic, increasing roughness usually lowers the water contact angle and strengthens hydrophilicity; for an already hydrophobic surface, roughness usually raises the contact angle and strengthens hydrophobicity [34]. Furthermore, the hydrated layer explains why most fouling is reversible and can be removed through physical cleaning, with only 4.5–9.6% resulting in irreversible fouling. Besides improving hydrophilicity through NPs addition, PVA, a hydrophilic polymer, likely gives the PVA-NP coated membrane the highest flux recovery ratio, as it may create a less rough surface with higher hydrophilicity, thereby decreasing irreversible fouling [32]. BSA is used to examine the anti-fouling

property as BSA acts as model protein and gives a model to understand the surface interactions with the membrane [35]. Unwanted protein and microbial adsorption on material surfaces, or biofouling, is a significant problem in water treatment applications and the membranes developed in this study try to address that problem.

3.5. Heavy metal removal performance of the fabricated membranes

The pH dependency of heavy metal removal efficiency by the modified membranes was examined using 1000 µg/l aqueous solutions of anionic species As(III), Cr(VI), and cationic species Pb(II), Cd(II). The study aimed to identify the pH range where all four metals can be simultaneously removed with maximum effectiveness. All experiments showed that the in-situ NP-grown membrane achieved the highest removal percentages for all four heavy metals within the observed pH range. Besides electrostatic interactions, adsorption is proposed to occur through abundant surface -OH groups that can form inner-sphere or outer-sphere complexes. These groups can accept or donate H⁺ ions, depending on pH, to create reactive surface sites. At high pH, -OH groups deprotonate to form M-O⁻, which can interact more strongly with cationic species and may contribute to higher cation removal. At low pH, the groups become protonated as M-OH²⁺, allowing stronger attraction and complexation with anionic species. As seen in Fig. 9, the data reveal a common trend: the removal percentage decreases after pH~10 for all metals. The high zero-point charge pH (~ 13.05) of the NPs helps explain these results. Although Cd forms cationic species at pH< 10, it becomes neutral Cd(OH)₂ or anionic species Cd(OH)₃⁻ and Cd(OH)₄²⁻ beyond pH~10. Similarly, Pb exists as Pb²⁺ in water up to pH ~ 9, after which Pb(OH)₃⁻ becomes the dominant species [23], [33]. At higher pH, the net surface charge of the NPs turns negative, which may reduce physisorption for anionic species due to electrostatic repulsion.



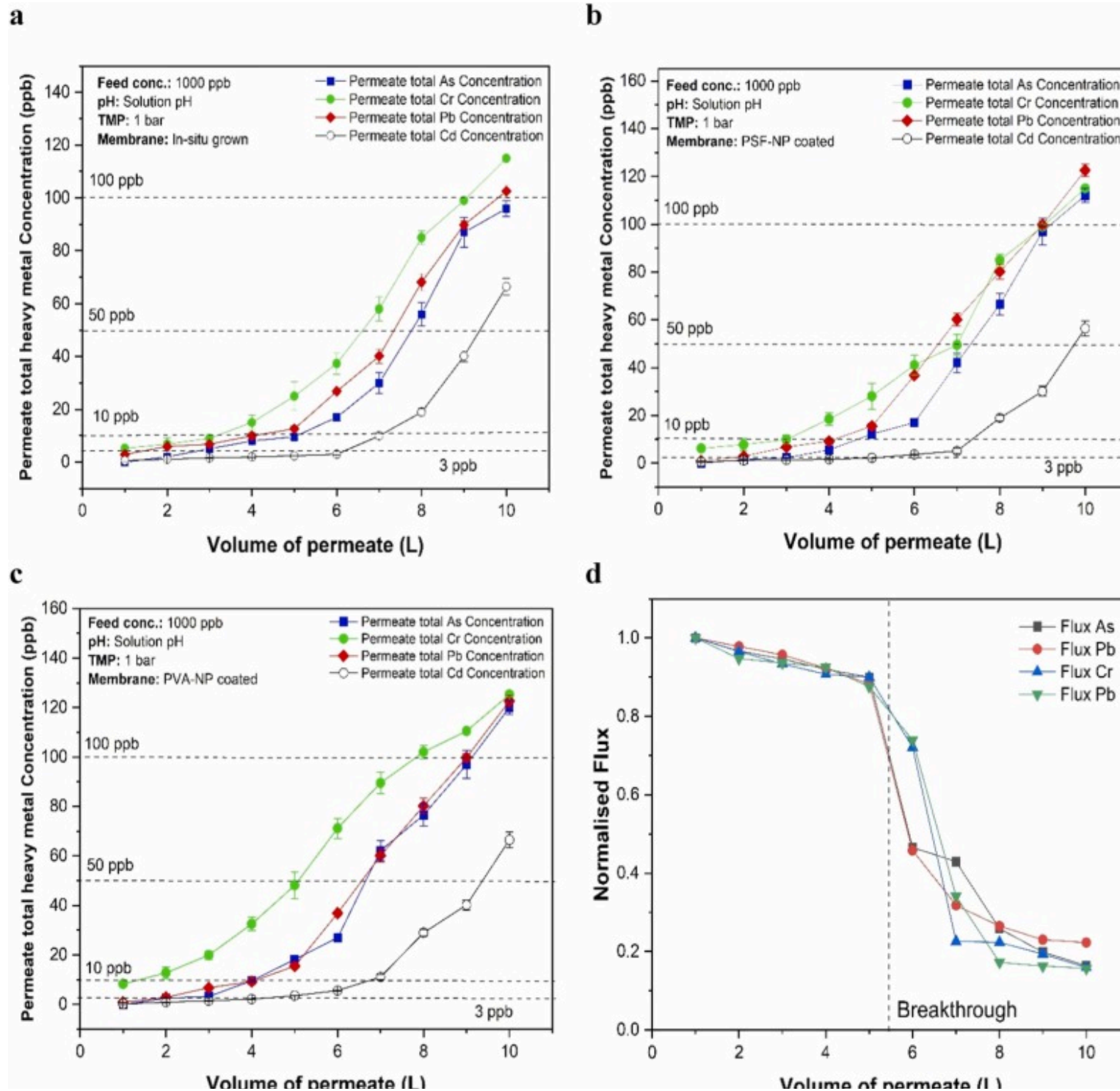
[Download: Download high-res image \(328KB\)](#)

[Download: Download full-size image](#)

Fig. 9. pH dependence of heavy-metal removal for the three membranes.

However, other mechanisms like inner and outer sphere complexation involving abundant -OH groups, accepting or donating H^+ ions to form $-OH^{2+}$ or $-O^-$, along with chemisorption affinity of the NPs, can account for the relatively small decrease in removal efficiency. This would not be possible if adsorption occurred only via physisorption. At lower pH, anionic species are adsorbed more due to electrostatic attraction with the positively charged adsorbent surface. This also explains the decline in removal percentages for Cd(II) and Pb(II) at very low pH. Overall, the experiments indicate that all membranes can remove over 90% of the heavy metals within the pH range of 4–10, with the in-situ NP-grown membrane showing the highest removal efficiency, followed by the PSF-P123-NP coated and PVA-NP coated membranes. The difference between removal percentages for the PVA-NP coated and PSF-P123-NP coated membranes can be attributed to a less uniform distribution of NPs in the coating process. Less apparent porosity of PVA-NP coated membranes can also mean that the NPs get stuck inside inaccessible pores, reducing the overall heavy metal removal performance of the membrane.

Long-term filtration studies for all four heavy metal species with 1000 $\mu\text{g/l}$ feed were performed to study the breakthrough performance of the membranes, where the breakthrough volume is the point of exhaustion for a membrane during a continuous filtration process. Fig. 10 shows the compilation of the long-term filtration studies.



[Download: Download high-res image \(569KB\)](#)

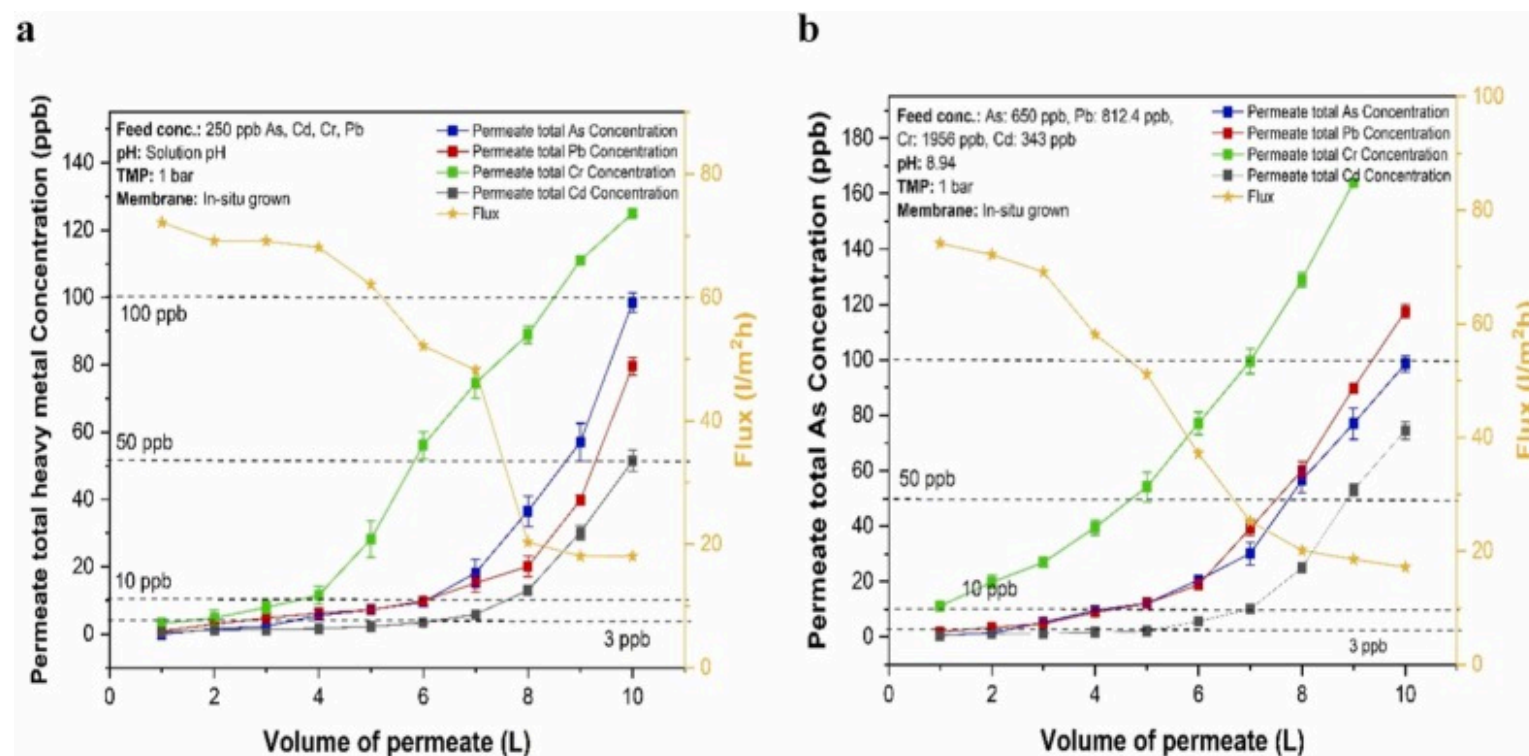
[Download: Download full-size image](#)

Fig. 10. a) long-term filtration by in-situ NP grown membrane, b) long-term filtration by PSF-P123-NP coated membrane, c) long-term filtration by PVA-NP coated membrane, d) normalised flux decline during long-term filtration.

The data suggests that the in-situ NP grown membrane has shown the overall best performance to remove individual heavy metals with breakthrough volumes of 5 L for As, 6 L for Cd, ~5 L for Pb and ~6 L for Cr. The other membranes have shown relatively lower breakthrough volumes, making the in-situ NP grown membrane the best option for long-term single heavy metal filtration. In long-term heavy metal removal, it was found that one tubular membrane with a length of 10 cm and an external diameter of 5 mm, giving a surface area of $\sim 0.001571 \text{ m}^2$, can remove 4 L of real arsenic contaminated ground water before reaching the breakthrough. By simple arithmetic, a 1 m^2 membrane should be able to purify $\sim 2500 \text{ L}$ of $1000 \mu\text{g/l}$ As(III) solution without regeneration before reaching the breakthrough while maintaining the same operating conditions. Similarly, $> 3000 \text{ L}$ of multi-metal solution consisting of As, Pb, Cd, and Cr at $250 \mu\text{g/l}$ concentration each can be purified without regeneration. PSF-P123-NP coated membrane has become a close second, probably due to the nature of the coating, which may reduce the surface area of the NPs, reducing active sites for adsorption. The comparatively poor performance by PVA-NP coated membrane may be attributed to the lack of porosity, and it is possible that disordered accumulation of NPs in the PVA matrix may have created dangling bonds, free radicals and unsaturated bonds, making a portion of the adsorbent inaccessible for adsorption [36]. Fig. 10d shows the normalised flux decline for the in-situ NP grown membrane to demonstrate the breakthrough volume and the decline in flux during the operation. During long-term operation, adsorbed heavy metal ions on the membrane surface and pores increase the adsorption resistance, resulting in the flux decline. Once the membrane is exhausted, the adsorption resistance increases dramatically, and the flux declines considerably [37].

It can be inferred from the studies so far that the in-situ NP grown membrane is relatively better at removing heavy metals than the other two membranes. The next set of experiments was conducted to study the multi-metal removal performance using the in-situ NP grown membrane. In Fig. 11a, the results of long-term filtration of a simulated multi-metal solution consisting of $250 \mu\text{g/l}$ of As, Pb, Cd, and Cr each are presented, along with the flux profile during operation. The membrane provided satisfactory removal for Cr up to 5 L, and for Pb, Cd, and As up to 6 L. This aligns with findings from batch adsorption experiments in Section

3.2. The lower adsorption performance for Cr likely results from the formation of complexes with abundant OH⁻ ions present in water under basic conditions. The OH⁻ ions probably attract to the six positive charges of Cr and shield it from adsorption sites [36]. After 6 L of filtration, the permeate flux significantly decreases, indicating membrane exhaustion. The same experiment was repeated using wastewater collected from a water body at East Kolkata Wetlands (22° 0' 27" N 88° 0' 27" E). The sample contained other heavy metals besides As, Pb, Cr, and Cd; components with insignificant concentrations were not considered for the study, as mentioned in Section 2.10.2. The in-situ NP grown membrane successfully removed Cd up to 5 L, and Pb, Cr, and As up to 4 L. The permeate flux gradually decreased and declined heavily as the membrane reached the breakthrough volume. From these results, it may be concluded that the in-situ NP grown membrane is suitable for practical multi-metal removal scenarios.



[Download: Download high-res image \(301KB\)](#)

[Download: Download full-size image](#)

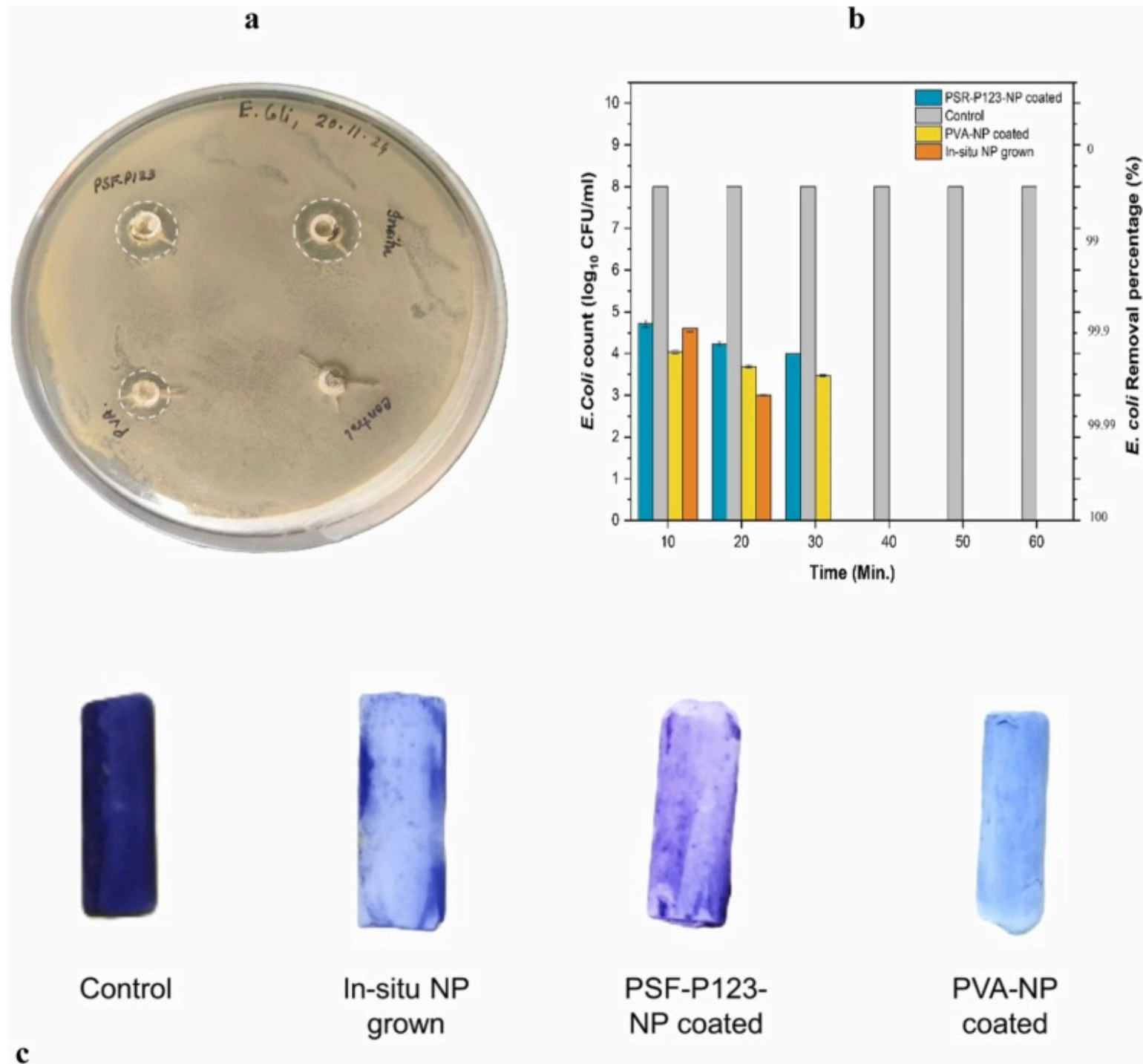
Fig. 11. a) simulated multi-metal aqueous solution long-term filtration, b) multi-metal contaminated real water sample long-term filtration.

3.6. Membrane anti-microbial performance

The anti-microbial performance of the nanocomposite membranes was studied through four different aspects: disc diffusion assay, solid surface antibacterial activity, resistance to biofilm formation and continuous filtration of bacteria-laden water to holistically study the different use-case scenarios of practical water treatment.

3.6.1. Disc diffusion assay

In many studies, different nanoparticles have been shown to inhibit microbial growth, whereas for nanocomposites, the release and diffusion of antimicrobial components decrease significantly. Therefore, the disc diffusion assay in this work was performed to see the potency of the nanocomposites, rather than the nanoparticles, to inhibit microbial growth. The result of the disc diffusion assay of the nanocomposite membranes against *E. coli* is shown in [Fig. 12a](#), and it indicates clear zones surrounding the nanocomposite membranes, which qualitatively evaluate their potency to inhibit or kill bacteria. The in-situ NP grown membrane showed a maximum zone of inhibition (24 mm), marking significant potential to inhibit or kill bacteria, followed by PSF-P123-NP coated ceramic membrane (22 mm) and PVA-NP coated membrane (16 mm).



[Download: Download high-res image \(321KB\)](#)

[Download: Download full-size image](#)

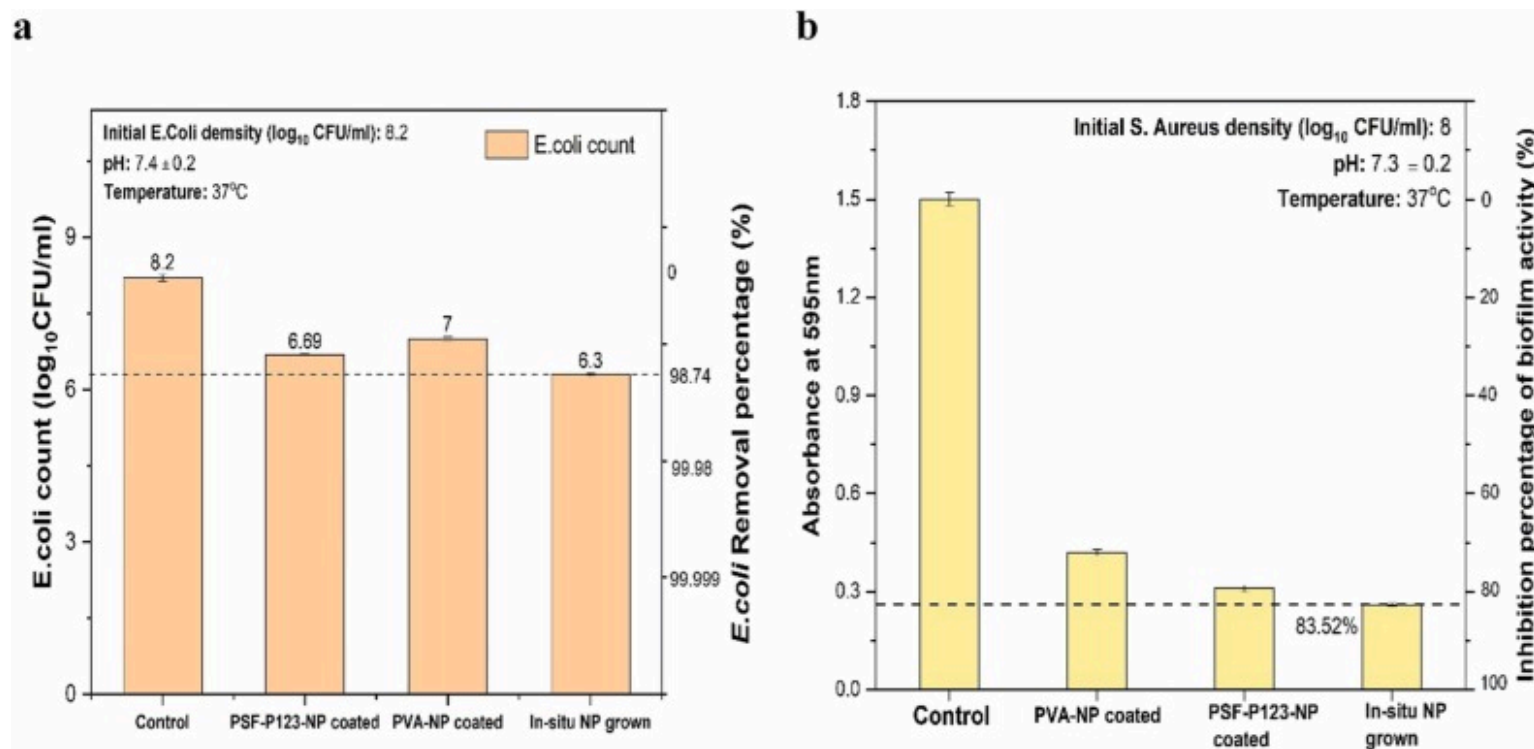
Fig. 12. a) disc diffusion assay, b) *E. coli* filtration experiment, c) biofilm inhibition study, higher staining indicated higher biofilm formation (qualitative).

Pristine ceramic membranes served as a control in this study, and no discernible antibacterial potency was observed, which indicated that unmodified ceramic membranes do not have intrinsic antibacterial efficiency. Bacterial inhibition in the disc diffusion process is due to a concentration gradient with the highest concentration near the disk, gradually decreasing as the distance from the disk increases. In the in-situ NP grown membrane, the nanoparticles are formed either on the surface or within the pores, thus contributing to better local interaction upon diffusion of the antibacterial agents near the disc. The presence of Mg in the ternary metal oxide in the nanocomposite is probably the reason for generating reactive oxygen species (ROS) and superoxide species, contributing to the inhibition of bacterial growth. Mg, an alkaline earth metal, easily donates 2 electrons, which helps in ROS generation. Previously established studies have shown that MgO can generate reactive oxygen species [38], [39]. Moreover, element doping-induced lattice distortion and defect formation, such as doping with Al and Ti in this case, can promote ROS generation and enhance the antibacterial properties of MgO nanomaterials [39]. Similar conclusions have been found in previous works [40]. The peptidoglycan cell layer, which consists of sugar and amino acids, gets disrupted by contact with the NP, resulting in protein and carbohydrate leakage [40]. The PSF-P123-NP-coated and PVA-NP-coated membranes showed smaller zone inhibition in the disc diffusion assay. One of the probable reasons behind a smaller zone of inhibition may be the non-uniform dispersion of NPs in polymer blends, which may reduce the amount of anti-microbial agents. The polymer-nanocomposite blend may also restrict the mobility of nanoparticles due to polymer entrapment, resulting in a smaller zone of inhibition. Although PVA allows better swelling and release of nanoparticles, agglomeration of nanoparticles and strong matrix interaction may still be the reason for hindrance in diffusion.

3.6.2. Solid surface anti-microbial activity

The disc diffusion assay is dependent on the concentration gradient of the antibacterial agents, and the inhibition of growth observed does not provide information about whether the antibacterial agent is bactericidal or bacteriostatic [41]. A solid surface antibacterial test was performed to overcome the limitation and quantify the reduction of bacterial cells. This assay was designed to allow the NPs to directly interact with the microbial cells during incubation.

Solid surface antibacterial tests of the modified and pristine ceramic membranes were conducted against *E. coli* with a defined contact time of 24 h. Bacterial cells were incubated directly on the membrane surface, followed by CFU counts to quantify the results. The findings are shown in [Fig. 13a](#), where the in-situ NP grown membrane achieved an antibacterial efficiency of 98%, reducing the CFU/ml count from an initial 1×10^8 CFU/ml to 2×10^6 after 24 h of incubation. The bactericidal mechanism primarily involves the interaction of bacterial cell surfaces with the NPs, leading to disruption. *E. coli* is a Gram-negative bacterium with a thin peptidoglycan layer surrounded by an outer lipopolysaccharide (LPS) layer. The LPS layer imparts a negative charge to the cell wall. The surface of NPs remains highly positively charged across most of the pH range, with a zero-point charge pH value of ~ 13 . These opposite charges facilitate effective interactions between the NPs and bacterial cells, causing cell wall damage. The thin peptidoglycan layer of *E. coli* further aids the NPs in penetrating the cell to enhance the killing process. The PSF-P123-NP coated membrane demonstrated an antibacterial efficiency of 95%, with a final CFU/ml count of 5×10^6 . The somewhat lower performance likely results from reduced contact area due to NPs being embedded in the polymer matrix. Nonetheless, the antimicrobial activity of the NPs is probably due to the release of ROS from the NPs at the bacteria-ceramic membrane interface, leading to cellular damage and death. PVA-nanocomposite blend ceramic membranes exhibited an antimicrobial efficiency of 90%, with a CFU/ml count of 1×10^7 . The results indicated that the PVA-NP coated membrane has lower antibacterial potential compared to in-situ coated and PSF nanocomposite membranes. Since the bactericidal activity mainly relies on the abundance of -OH groups in the NPs, increased hydrogen bonding with PVA may reduce ROS availability, thereby decreasing antimicrobial effectiveness. [\[42\]](#).



[Download: Download high-res image \(226KB\)](#)

[Download: Download full-size image](#)

Fig. 13. a) Solid surface anti-microbial activity (quantitative), b) Biofilm inhibition activity (quantitative).

3.6.3. Biofilm inhibition activity

Biofilm forms when bacteria multiply and attach to each other or to living and non-living surfaces using an extracellular matrix. This type of attachment is often observed on the membranes of water treatment units, which can interfere with filter shelf life. Therefore, studying the ability of a membrane to prevent bacterial biofilm formation on its surface is necessary for better practical application. The current study examined the adhesion and colonisation of *S. aureus* on the surface of ceramic membranes with different modifications. Both qualitative (Fig. 12c) and quantitative (Fig. 13b) analyses were performed after three days of incubation. The qualitative assessment relies on visual evaluation of the stained biofilm on the modified and pristine ceramic membranes. It is based on the intensity of dye absorption by these membranes. The in-situ-NP-grown

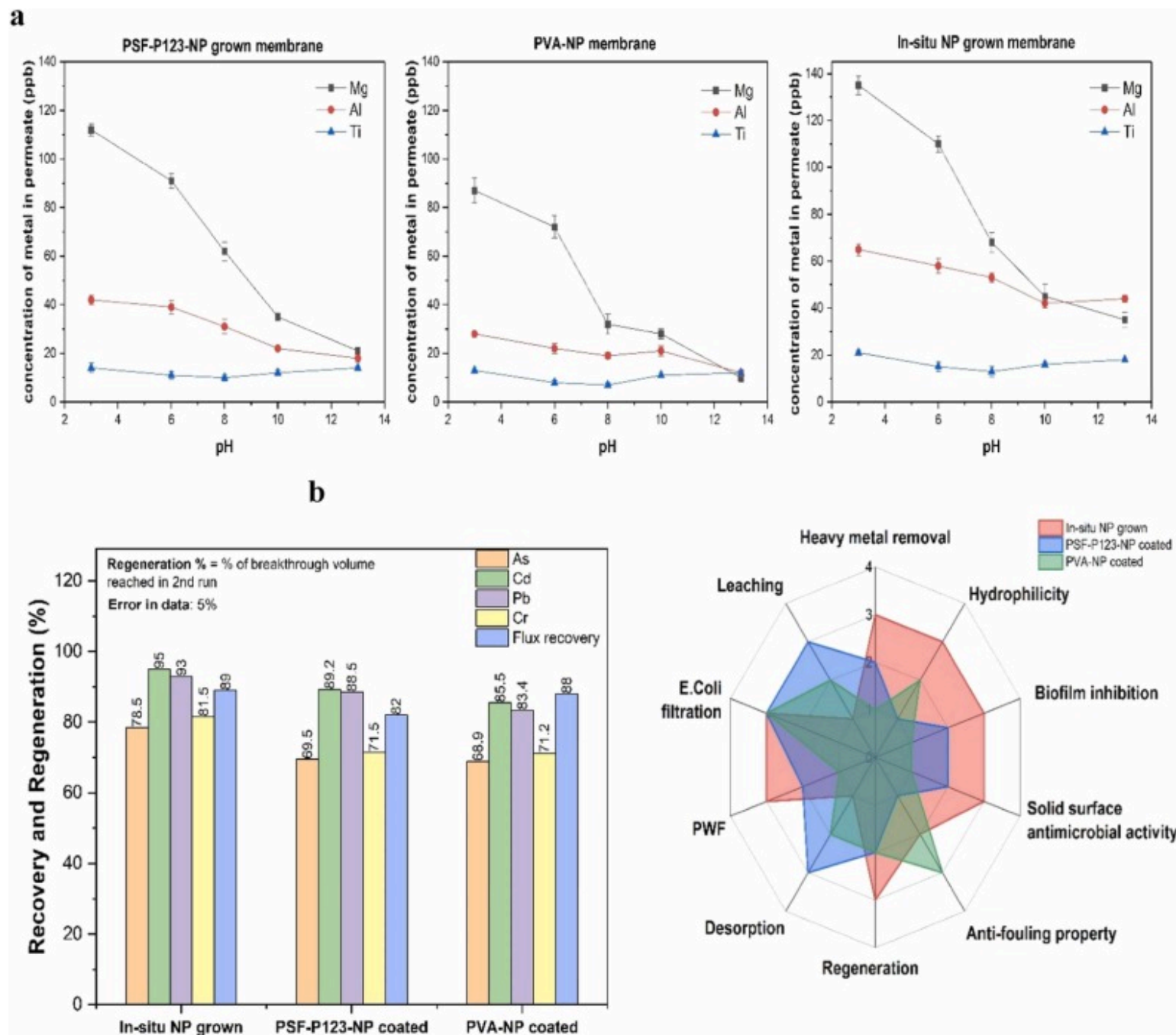
membrane showed minimal retention of crystal violet dye upon staining, indicating the least biofilm formation on its surface. PSF-P123-NP-coated and PVA-NP-coated membranes showed moderate dye retention, indicating some biofilm formation. All three modified ceramic membranes were compared qualitatively against the pristine ceramic membrane, which showed intense staining and indicated a substantial amount of biofilm. Quantitative analysis revealed a biofilm inhibition rate of 83.52% for the in-situ NP grown membrane, followed by the PSF-P123-NP coated membrane with 80%, and the PVA-NP coated membrane with the least inhibition at 75.29%. The pristine ceramic membrane showed no antibiofilm activity. The proposed mechanism behind the antibiofilm activity of the modified membranes is likely due to the positive charge on the nanocomposites, which produces electrostatic repulsion against Gram-positive bacteria, preventing biofilm formation. Since Mg-Al-Ti oxide NPs have a zero-point charge pH of approximately 13, the net surface charge will be highly positive at around pH 7, where the experiment was conducted. The highest biofilm inhibition observed with the in-situ NP grown membrane is attributed to its larger surface area for nanoparticle exposure, the release of metal ions, and the abundance of hydroxyl groups acting as reactive oxygen species. These factors disrupt the cellular membrane, killing bacteria in the planktonic stage before biofilm initiation [44].

3.6.4. *E. coli* filtration experiment

Finally, to gauge the real application potential of the anti-microbial activity, an *E. coli* filtration experiment was performed using a cell density of 1×10^8 over 60 60-minute duration. The observations are recorded as the percentage removal of *E. coli* by the modified ceramic membranes (Fig. 12b). All three modified ceramic membranes could remove 100% bacteria during a 60-minute filtration experiment. The samples were drawn in intervals of 10 min to maintain the accuracy of the performance of the modified membranes. The in-situ NP grown membrane achieved a 100% *E. coli* removal efficiency within 30 mins of the filtration experiment. No bacterial colony was formed after plating on agar media. This outcome surpasses previously reported works, where researchers attained 99% eradication efficiency for *E. coli* [43]. PSF-P123-NP-coated membrane and PVA-NP-coated membrane achieved 100% efficiency within 40 min. The possible mechanism behind the 100% antibacterial efficiency exhibited by the modified ceramic membranes can be stated as the combined effect of physical entrapment of *E. coli* cells by the membrane, followed by cell death. The nanocomposite incorporated in the ceramic membrane comes in contact with the *E. coli* cell surface, inducing a contact-based killing of the cells. The observations from the solid surface antibacterial test support this suggested mechanism. From the observations of the present study, it can be concluded that the modified ceramic membranes can remove pathogens with 100% efficiency and are perfect for deployment in real-world applications.

3.7. Leaching, desorption, and regeneration

Leaching is one of the most important parameters concerning the real usage scenario of nanocomposite membranes in water treatment. The arsenic filtration experiment at different pH levels was used to study the leaching behaviour of the membrane at different pH levels. Fig. 14a shows the Mg, Al and Ti concentrations in the permeate during As(III) removal at different pH levels. From the literature, it can be found that the highest allowable limits of Mg and Al in water are 30–100 mg/l, 30–200 µg/l, respectively; Ti has been reported to have a safe limit of 0.5–15 µg/l, but the studies show different variations in the result for Ti [5], [44]. For all the membranes, the general trend is that the concentration of all the constituent metals is high at the lower pH level, as in this range, the oxides can dissolve in water. As pH increases, the leaching decreases drastically, and all the values fall under the safe drinking water level. It should be noted that the increase in the concentration of Mg at lower pH values denotes that the removal of As is largely due to -OH groups bound to Mg. Moreover, the in-situ NP grown membrane exhibits higher leaching of all three constituent metals than the two coated membranes. Although the concentration values are within the safe range, higher leaching may be considered a drawback of the membrane.



Download: [Download high-res image \(417KB\)](#)

[Download: Download full-size image](#)

Fig. 14. a) Leaching of constituent metals of the adsorbent, b) Membrane regeneration and flux recovery, c) radar diagram for comparing different features of the membranes.

Desorption was studied to understand if the captured heavy metals can return to the environment after discarding the membranes. It was found that in the whole pH range, in-situ NP grown membrane, PSF-P123-NP coated membrane, and the PVA-NP coated membrane showed desorption of 5–10%, 3–5% and 5–8%, respectively, after long-term heavy metal filtration. Again, in this case, the in-situ NP grown membrane shows a higher desorption level due to more contact of the spent NPs with water, while the coating on the other two membranes limits the desorption.

The less hydrophilic nature of the PSF-P123-NP coated membrane may be limiting the desorption even more. This low desorption means that bonds are formed during the adsorption process, suggesting both physisorption and chemisorption are contributing to the removal of heavy metals.

Membrane regeneration was done using NaOH solution and aqueous solution of MgO (pH~10.3), followed by neutralisation by HCl. The results are presented in [Fig. 14b](#) for regeneration and flux recovery after long-term filtration of all four heavy metals. After that, the same feed from the multi-metal removal experiment was used to evaluate regeneration performance. Regeneration percentage has been represented as the percentage of breakthrough volume reached in the second run. The regeneration percentages are different for different heavy metal species, but it may be inferred that a higher regeneration percentage was achieved for spent membranes adsorbing cationic species, while the regeneration is poorer in the case of anionic species. This also suggests that physisorption is more dominant while adsorbing Cd and Pb species at the solution pH (~7), giving more reversibility to the process, while As, Cr adsorption proceeds mostly through complexation, making the regeneration much less. Flux recovery was similar for all the membranes, and the highest flux recovery of 89% could be achieved for the in-situ NP grown membrane.

3.8. Comparative analysis

A radar diagram was prepared ([Fig. 14c](#)) consisting of ten different features (heavy metal removal, pure water flux, anti-fouling property, hydrophilicity, leaching, desorption, regeneration, biofilm inhibition, solid state anti-microbial activity) of the membranes fabricated in this work to get a compact idea of the abilities of the membranes. Membranes were ranked from 1 to 3

based on their performance, where 3 is the best, and 1 is the worst. The in-situ NP grown membrane comes out as the best-performing membrane, ranking the best in seven features, while the PSF-P123-NP-coated membrane showed the best desorption and leaching performance. PVA-NP coated membrane showed the best anti-fouling performance while performing satisfactorily in other aspects. All the membranes performed equally well during *E. Coli* filtration and could remove > 99.99% bacteria after 60 min. Table 1 contains a list of ceramic composite membranes in the literature and the performance of the membranes prepared in this work. However, such comparative work suffers from a few limitations, like, all the fabrication methods could ideally be optimised for the best membrane preparation, the NP loading in the in-situ NP grown membrane is measured through approximation.

4. Conclusion

Mg-Al-Ti ternary oxide nanoparticles (NPs) were prepared through an evaporation-induced self-assembly method, which can simultaneously remove cationic and anionic heavy metal species (As, Pb, Cd, Cr) from water across a wide pH range of 3–10. Three novel membranes were fabricated based on tubular ceramic support membranes (length: 10 cm, external diameter: 5 mm, average pore size: 1.5 μm) made of alumina. PVA-NP coated membranes and PSF-P123-NP coated membranes were created to achieve surface-level modification, while in-situ NP grown membranes caused matrix-level modification. The in-situ NP grown membrane can treat ~6 L of 1 mg/l single-metal contaminated water and ~4 L of real multi-metal contaminated water before requiring regeneration. The NPs significantly enhanced the membranes' hydrophilicity from 69.83° (unmodified) to 25.06° for the in-situ NP grown membrane, and gave the membranes anti-fouling properties, with a maximum of 95.47% flux recovery for the PVA-NP coated membrane while permeating 1000 mg/l BSA solution. The membranes also exhibited anti-microbial properties; the in-situ NP grown membrane showed the largest inhibition zone in disc diffusion tests (24 mm), the highest effectiveness in solid-surface anti-microbial activity (98%), and biofilm inhibition (83.52%), with PSF-P123-NP and PVA-NP coated membranes ranking second and third. All membranes achieved over 99.99% removal of *E. coli* during continuous filtration. With minimal desorption (3–5%), PSF-P123-NP membranes are the safest for disposal, whereas the in-situ NP grown membranes exhibited the highest leaching of NP constituent metals. The in-situ NP grown membrane demonstrated the greatest regeneration efficiency, particularly after adsorption of cationic species. This study compares three different novel ceramic nanocomposite membranes, highlighting the effectiveness of different fabrication strategies. The most important takeaway could be that, in future, ceramic composite membranes can be developed keeping in mind the specific property the application needs. If the application needs more importance on anti-fouling property, membranes can be prepared using PVA or hydrophilic polymer-based coating. The

performance of all three membranes indicates their potential for practical water treatment applications, tailored to their specific features and the requirements of different uses, helping to address the shortage of multifunctional nanocomposite ceramic membranes.

CRedit authorship contribution statement

Umme Habiba: Methodology, Formal analysis, Data curation. **Somakraj Banerjee:** Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Ranjana Das:** Writing – review & editing, Supervision, Resources, Project administration, Investigation. **Aishwarya Das:** Methodology, Formal analysis, Data curation. **Chiranjib Bhattacharjee:** Visualization, Supervision, Software, Resources, Project administration, Investigation, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors sincerely thank the Department of Science and Technology, Govt. of India for providing sponsorship for this research work through the project entitled “Centre for Technological Excellence in Water Purification (CTEWP)” under Water Treatment Initiative (WTI) (vide sanction letter no DST/TM/WTI/WIC/2K17/84(c) dated 21.02.2019). Authors also acknowledge Science & Technology and Biotechnology, Govt. of West Bengal for financial assistance (vide sanction letter no. 2188(Sanc.)/STBT-11012(25)/8/2024-ST-SEC) dated 10.02.2024).

[Recommended articles](#)

Data availability

Data will be made available on request.

References

- [1] Y. Xu, X. Zhao, R. Chang, H. Qu, J. Xu, J. Ma
Designing heterogeneous MOF-on-MOF membrane with hierarchical pores for effective water treatment
J. Memb. Sci., 658 (2022), Article 120737, [10.1016/J.MEMSCI.2022.120737](https://doi.org/10.1016/J.MEMSCI.2022.120737) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [2] M.E. Argun, S. Dursun
A new approach to modification of natural adsorbent for heavy metal adsorption
Bioresour. Technol., 99 (2008), pp. 2516-2527, [10.1016/J.BIORTECH.2007.04.037](https://doi.org/10.1016/J.BIORTECH.2007.04.037) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [3] Guidelines for Water Quality Monitoring Central Pollution Control Board Parivesh Bhawan East Arjun Nagar, Delhi-32, (n.d.).
- [4] N.A.A. Qasem, R.H. Mohammed, D.U. Lawal
Removal of heavy metal ions from wastewater: a comprehensive and critical review
4
Npj Clean. Water, 4 (1) (2021), pp. 1-15, [10.1038/s41545-021-00127-0](https://doi.org/10.1038/s41545-021-00127-0) ↗
[Google Scholar](#) ↗
- [5] M. Bhavan, B. Shah, Z. Marg, BUREAU OF INDIAN STANDARDS, Bureau of Indian Standards (BIS) (2012).
[Google Scholar](#) ↗
- [6] Y. Zeng, C. Yang, J. Zhang, W. Pu
Feasibility investigation of oily wastewater treatment by combination of zinc and PAM in coagulation/flocculation
J. Hazard. Mater., 147 (2007), pp. 991-996, [10.1016/j.jhazmat.2007.01.129](https://doi.org/10.1016/j.jhazmat.2007.01.129) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [7] I. Jacukowicz-Sobala, A. Ciechanowska, E. Kociółek-Balawejder, A. Gibas, A. Zakrzewski

Photocatalytically-assisted oxidative adsorption of As(III) using sustainable multifunctional composite material – Cu₂O doped anion exchanger

J. Hazard. Mater., 431 (2022), [10.1016/J.JHAZMAT.2022.128529](https://doi.org/10.1016/J.JHAZMAT.2022.128529) ↗

[Google Scholar](#) ↗

- [8] J.N. Hakizimana, B. Gourich, M. Chafi, Y. Stiriba, C. Vial, P. Drogui, J. Naja
Electrocoagulation process in water treatment: a review of electrocoagulation modeling approaches

Desalination, 404 (2017), pp. 1-21, [10.1016/j.desal.2016.10.011](https://doi.org/10.1016/j.desal.2016.10.011) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [9] B. Van der Bruggen, M. Mänttari, M. Nyström
Drawbacks of applying nanofiltration and how to avoid them: a review

Sep. Purif. Technol., 63 (2008), pp. 251-263, [10.1016/J.SEPPUR.2008.05.010](https://doi.org/10.1016/J.SEPPUR.2008.05.010) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [10] W.S. Choi, H.-J. Lee, W.S. Choi, H.-J. Lee
Nanostructured materials for water purification: adsorption of heavy metal ions and organic dyes

Polymers, 14 (2022), p. 2183, [10.3390/POLYM14112183](https://doi.org/10.3390/POLYM14112183) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [11] F. Yang, G. Ding, J. Wang, Z. Liang, B. Gao, M. Dou, C. Xu, S. Li
Self-cleaning, antimicrobial, and antifouling membrane via integrating mesoporous graphitic carbon nitride into polyvinylidene fluoride

J. Memb. Sci., 606 (2020), Article 118146, [10.1016/J.MEMSCI.2020.118146](https://doi.org/10.1016/J.MEMSCI.2020.118146) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [12] S. Banerjee, A. Das, R. Das, C. Bhattacharjee
Mesoporous Mg–Al–Ti composite oxide incorporated mixed matrix ultrafiltration membrane with superior multi-heavy metal removal capacity, anti-fouling & anti-microbial property

J. Memb. Sci., 703 (2024), Article 122836, [10.1016/J.MEMSCI.2024.122836](https://doi.org/10.1016/J.MEMSCI.2024.122836) ↗

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [13] S. Wu, W. Shi, L. Cui, C. Xu
Enhancing contaminant rejection efficiency with ZIF-8 molecular sieving in sustainable mixed matrix membranes

Chem. Eng. J., 482 (2024), Article 148954, [10.1016/J.CEJ.2024.148954 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [14] M. Saleh, S. Gonca, Z. Isik, Y. Ozay, E. Harputlu, S. Ozdemir, M. Yalvac, K. Ocakoglu, N. Dizge
Preparation of ZnO nanorods or SiO₂ nanoparticles grafted onto basalt ceramic membrane and the use for E. coli removal from water

Ceram. Int., 47 (2021), pp. 27710-27717, [10.1016/J.CERAMINT.2021.06.196 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [15] D. Menne, C. Üzümlü, A. Koppelman, J.E. Wong, C. van Foeken, F. Borre, L. Dähne, T. Laakso, A. Pihlajamäki, M. Wessling
Regenerable polymer/ceramic hybrid nanofiltration membrane based on polyelectrolyte assembly by layer-by-layer technique

J. Memb. Sci., 520 (2016), pp. 924-932, [10.1016/J.MEMSCI.2016.08.048 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [16] E. Maaskant, P. de Wit, N.E. Benes
Direct interfacial polymerization onto thin ceramic hollow fibers

J. Memb. Sci., 550 (2018), pp. 296-301, [10.1016/J.MEMSCI.2018.01.009 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [17] J.B.S. Hamm, A. Ambrosi, L.D. Pollo, N.R. Marcilio, I.C. Tessaro
Thin polymer layer-covered porous alumina tubular membranes prepared via a dip-coating/phase-inversion process

Mater. Chem. Phys., 265 (2021), Article 124511, [10.1016/J.MATCHEMPHYS.2021.124511 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [18] H. Bai, L. Liang, P. Cao, H. Zhang, S. Chen, H. Yu, X. Quan
MgAl₂O₄ incorporated catalytic ceramic membrane for catalytic ozonation of organic pollutants
Appl. Catal. B, 343 (2024), Article 123527, [10.1016/j.apcatb.2023.123527](https://doi.org/10.1016/j.apcatb.2023.123527) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [19] S. Banerjee, R. Das, C. Bhattacharjee
Exceptional arsenic removal using novel mesoporous Mg-Al-Ti ternary composite oxide nanoparticles prepared via self-assembly templating method
Sep. Purif. Technol., 326 (2023), Article 124795, [10.1016/j.seppur.2023.124795](https://doi.org/10.1016/j.seppur.2023.124795) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [20] S. Roy, S. Majumdar, G.C. Sahoo, S. Bhowmick, A.K. Kundu, P. Mondal
Removal of As(V), Cr(VI) and Cu(II) using novel amine functionalized composite nanofiltration membranes fabricated on ceramic tubular substrate
J. Hazard. Mater., 399 (2020), Article 122841, [10.1016/j.jhazmat.2020.122841](https://doi.org/10.1016/j.jhazmat.2020.122841) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [21] N. Barati, M.M. Husein, J. Azaiez
Modifying ceramic membranes with in situ grown iron oxide nanoparticles and their use for oily water treatment
J. Memb. Sci., 617 (2021), Article 118641, [10.1016/j.memsci.2020.118641](https://doi.org/10.1016/j.memsci.2020.118641) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [22] A.K. Mallik, M.A. Moktadir, M.A. Rahman, M. Shahruzzaman, M.M. Rahman
Progress in surface-modified silicas for Cr(VI) adsorption: a review
J. Hazard. Mater., 423 (2022), Article 127041, [10.1016/j.jhazmat.2021.127041](https://doi.org/10.1016/j.jhazmat.2021.127041) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [23] D. Xu, X.L. Tan, C.L. Chen
X.K. Wang, Adsorption of Pb(II) from aqueous solution to MX-80 bentonite: effect of pH, ionic strength, foreign ions and temperature

Appl. Clay Sci., 41 (2008), pp. 37-46, [10.1016/J.CLAY.2007.09.004](https://doi.org/10.1016/J.CLAY.2007.09.004) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [24] Test Method for Water Absorption, Bulk Density, Apparent Porosity, and Apparent Specific Gravity of Fired Whiteware Products, Ceramic Tiles, and Glass Tiles, (2014). <https://doi.org/10.1520/C0373-14> ↗

[Google Scholar](#) ↗

- [25] M. Saleh, Z. Isik, P. Belibagli, H. Arslan, S. Gonca, S. Özdemir, N. Kudaibergenov, A. Khataee, N. Dizge
Fabrication of Ag nanoparticles coated leonardite basalt ceramic membrane with improved antimicrobial properties for DNA cleavage, E. coli removal and antibiofilm effects

J. Ind. Eng. Chem., 128 (2023), pp. 532-541, [10.1016/J.JIEC.2023.08.018](https://doi.org/10.1016/J.JIEC.2023.08.018) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [26] J. Nowak, C.D. Cruz, J. Palmer, G.C. Fletcher, S. Flint
Biofilm formation of the L. monocytogenes strain 15G01 is influenced by changes in environmental conditions

J. Microbiol. Methods, 119 (2015), pp. 189-195, [10.1016/J.MIMET.2015.10.022](https://doi.org/10.1016/J.MIMET.2015.10.022) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [27] R.J. Smith, E.B. Rampe, B.H.N. Horgan, E. Dehouck
Deriving amorphous component abundance and composition of rocks and sediments on earth and mars

J. Geophys. Res. Planets, 123 (2018), pp. 2485-2505, [10.1029/2018JE005612](https://doi.org/10.1029/2018JE005612) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [28] M. Valiskó, D. Boda, D. Gillespie
Selective adsorption of ions with different diameter and valence at highly charged interfaces

J. Phys. Chem. C., 111 (2007), pp. 15575-15585, [10.1021/JP073703C](https://doi.org/10.1021/JP073703C) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [29] Y. Wang, R. Wang, X. Ding, H. He

Pore-size-dependent adsorption of HCrO_4^- and $\text{Cr}_2\text{O}_7^{2-}$ towards sustainable supported Cr(III) catalysts through in-situ reduction

Sep. Purif. Technol., 361 (2025), Article 131601, [10.1016/J.SEPPUR.2025.131601](https://doi.org/10.1016/J.SEPPUR.2025.131601) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [30] J. Mansouri, E. Yapit, V. Chen
Polysulfone filtration membranes with isoporous structures prepared by a combination of dip-coating and breath figure approach

J. Memb. Sci., 444 (2013), pp. 237-251, [10.1016/J.MEMSCI.2013.05.022](https://doi.org/10.1016/J.MEMSCI.2013.05.022) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [31] M. Najim Obaid, O.H. Sabr, B. Jawad Kadhim
The effect of MgO nanoparticle on PVA/PEG-based membranes for potential application in wound healing

J. Biomater. Sci. Polym. Ed., 35 (2024), pp. 1963-1977, [10.1080/09205063.2024.2364526](https://doi.org/10.1080/09205063.2024.2364526) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [32] B. Liu, J. Zhang, H. Guo
Research progress of polyvinyl alcohol water-resistant film materials

Membranes, 12 (2022), p. 347, [10.3390/MEMBRANES12030347](https://doi.org/10.3390/MEMBRANES12030347) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [33] S. Sasaki, S. Omata, T. Murakami, N. Nagasawa, M. Taguchi, A. Suzuki
Effect of gamma ray irradiation on friction property of poly(Vinyl alcohol) cast-drying on freeze-thawed hybrid gel

Gels, 4 (2018), [10.3390/GELS4020030](https://doi.org/10.3390/GELS4020030) ↗

[Google Scholar](#) ↗

- [34] Q. Du, P. Zhou, Y. Pan, X. Qu, L. Liu, H. Yu, J. Hou
Influence of hydrophobicity and roughness on the wetting and flow resistance of water droplets on solid surface: a many-body dissipative particle dynamics study

Chem. Eng. Sci., 249 (2022), Article 117327, [10.1016/J.CES.2021.117327](https://doi.org/10.1016/J.CES.2021.117327) ↗

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [35] K. Jayanthi, P. Pazhanisamy, K. Parthipan, A. Thirunavukkarasu, P. Krishnan, K. Chithra, W. Shanthi, D. Satheesh
Synthesis, characterization, BSA adsorption and antifouling studies of N-cyclohexylacrylamide based hydrogels

Hybrid. Adv., 11 (2025), Article 100574, [10.1016/J.HYBADV.2025.100574 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [36] S.F. Bdewi, O.G. Abdullah, B.K. Aziz, A.A.R. Mutar
Synthesis, structural and optical characterization of MgO nanocrystalline embedded in PVA matrix

J. Inorg. Organomet. Polym. Mater., 26 (2016), pp. 326-334, [10.1007/S10904-015-0321-3/TABLES/2 ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [37] R. Mukherjee, P. Bhunia, S. De
Impact of graphene oxide on removal of heavy metals using mixed matrix membrane

Chem. Eng. J., 292 (2016), pp. 284-297, [10.1016/J.CEJ.2016.02.015 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [38] W.M. Girma, M.S. Aragie, B.A. Berehe, A.H. Assen
Biogenic synthesis and characterization of MgO nanoparticles using Verbascum sinaiticum: antibacterial, free radical, and reactive oxygen species scavenging activities

Mater. Adv., 6 (2025), pp. 4003-4015, [10.1039/D4MA01279H ↗](#)

[View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [39] J. Liu, B. Ouyang, W. Dun, P. Chen, T. Zhang, H. Dong, Y. Qing, W. Liu, Y. He, H. Wang
Revealing the oxidative stress mechanism induced by defect engineering of magnesium oxide nanoparticles under dark conditions

J. Environ. Chem. Eng., 13 (2025), Article 119192, [10.1016/J.JECE.2025.119192 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [40] P. Bhattacharya, A. Dey, S. Neogi

An insight into the mechanism of antibacterial activity by magnesium oxide nanoparticles

J. Mater. Chem. B, 9 (2021), pp. 5329-5339, [10.1039/D1TB00875G](https://doi.org/10.1039/D1TB00875G) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [41] J. Jorgensen, M.F.-C.I. Diseases
Antimicrobial susceptibility testing: general principles and contemporary practices

JSTOR]H Jorgensen, MJ FerraroClinical Infectious Diseases, JSTOR (n.d.) (1998)

<https://www.jstor.org/stable/4481507> ↗

(accessed June 26, 2025).

[Google Scholar](#) ↗

- [42] S. Phattarateera, M. Sangthongdee, T. Subsomboon, P. Threepopnatkul
Assessing antibacterial properties of polyvinyl alcohol/pregelatinized starch films for outbreak prevention

Ind. Crops Prod., 211 (2024), Article 118214, [10.1016/J.INDCROP.2024.118214](https://doi.org/10.1016/J.INDCROP.2024.118214) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [43] D. Marreiro, M. Saraniti, S. Aboud, S. Diana, R. Fauzan, E. Elfiana
Removing escherichia coli bacteria in river water using ceramic membrane from mixed clay and fly ash material

IOP Conf. Ser. Mater. Sci. Eng., 536 (2019), Article 012089, [10.1088/1757-899X/536/1/012089](https://doi.org/10.1088/1757-899X/536/1/012089) ↗

[Google Scholar](#) ↗

- [44] World Health Organization
Environmental Health Criteria 24 TITANIUM
INTERNATIONAL PROGRAMME ON CHEMICAL SAFETY (1982)

[Google Scholar](#) ↗

- [45] Y. Mei, J. Yang, R. Zhang, H. Li, Y. Guo
Enhanced antibacterial activity, dye rejection and anti-fouling performance of ZrO₂-SiO₂ composite ceramic membranes embedded by silver nanoparticles

J. Sol. -Gel Sci. Technol., 114 (2) (2025), pp. 413-429, [10.1007/S10971-025-06697-6](https://doi.org/10.1007/S10971-025-06697-6) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [46] W. Liu, D. Wang, R.A. Soomro, F. Fu, N. Qiao, Y. Yu, R. Wang, B. Xu
Ceramic supported attapulgite-graphene oxide composite membrane for efficient removal of heavy metal contamination

J. Memb. Sci., 591 (2019), Article 117323, [10.1016/J.MEMSCI.2019.117323](https://doi.org/10.1016/J.MEMSCI.2019.117323) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [47] S. Jana, A. Saikia, M.K. Purkait, K. Mohanty
Chitosan based ceramic ultrafiltration membrane: preparation, characterization and application to remove Hg(II) and As(III) using polymer enhanced ultrafiltration

Chem. Eng. J., 170 (2011), pp. 209-219, [10.1016/J.CEJ.2011.03.056](https://doi.org/10.1016/J.CEJ.2011.03.056) ↗

[Google Scholar](#) ↗

Cited by (0)

[View Abstract](#)

© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

